Closing Edge Loops at Quantization Time: Helix Constraints for T-Mesh Quantization

Research note — preprint, version 1, not peer reviewed

Marc Gardent*
Independent researcher
ORCID: 0009-0000-6466-3098
Contact: through the repository, `https://github.com/marcgardent/openclare`

18 October 2026

Abstract

Automatic quad remeshing often produces edge loops that wind round a part of the shape as helices instead of closing into rings. Bommes et al. [8] remove such *q-helices* after meshing; we prevent them where the integer structure of the mesh is decided, in the quantization of a T-mesh. A straight walk across the rectangles of a T-mesh that returns to its first side, a *turn*, has a *shift* that is an integer linear form in the arc lengths; at integer lengths it is how far the dual line following the same patches comes back beside itself. We add the constraints that close turns to a minimum-deviation quantizer [26], through a least-change projection, repair walks and rounds of constraint generation on the strips of the mesh, and keep the candidate with the fewest spiral edges. On twelve models at three budgets near 3,000 quads, the engine leaves 4.65% of the edges on spiral loops (16.7% without closing, in single runs of an earlier engine); on a human body, single runs give 1.6%, against 2.5% for the hand-made mesh. Reading the shift as translational holonomy of the flat metric, we show that it is the Burgers vector of the singularities on one side of the turn, that a dislocation joined by a staircase of arcs between two turns forbids closing both, and that on a closed sphere the shift of a turn has the parity of its length. Turned around, the conflicts bound from below the number of turns of a family that every quantization within given bounds leaves shifted: on the body, within the engine's bounds of separation, one to nine of the turns it holds, where it leaves 13 to 123 shifted; with its deviation cap and window of sizes as well, exactly one in two ways where it leaves 13 and 16, and in the others certified ranges from 1–12 to 9–90. Choosing the turns to close is NP-hard. The algebraic cores of the proofs are checked in Lean.

1 Introduction

Field-aligned quad remeshing has reached a level where element shape and alignment are rarely the limiting factor [10, 11]. What remains is structure: the way edge loops run across the surface. Artists expect a loop round an arm, a leg or a neck to be a ring. Automatic methods often produce a helix instead: the loop goes round, comes back one or a few rows further, and winds along the whole part. The problem is old and well identified: Bommes et al. showed in 2011 that helical configurations are a main reason why the base complex of an otherwise good quad mesh stays dense [8].

*This note is distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). The code of openclare and the Lean proofs are under the GPL-3.0-or-later of the repository. The note describes the engine and the Lean proofs at commit `3b890ecc8080` of `https://github.com/marcgardent/openclare`.

This note documents an approach developed in the open-source remesher openclare, and positions it against the literature we could check. What we claim as new, as far as we found, is listed first; the known results it builds on are attributed where they are used.

1. *Turns* of a T-mesh and their *shift*, a linear form in the arc lengths of the T-mesh that gives, at integer lengths and within the corridor of the turn, the pitch of the dual line that follows it (Section 3). That a translational period of a flat metric is linear in the lengths is classical; its form on a T-mesh, as a constraint of the quantization, is what is new.

2. Closing turns within T-mesh quantization: a least-change projection of the target lengths, an integer search, repair walks judged on the turns, rounds of constraint generation on the strips of the mesh, and a choice among candidates by the share of spiral edges, with safeguards on folds, distortion and budget (Section 4); measured on a bench that holds the result against open-source remeshers and a mesh modelled by hand (Section 6).

3. A share of spiral edges that tells helices of pitch up to four from closed, open and border loops (Section 5); loop counts and loop simplicity scores exist [38, 43], but they do not separate helices from open loops.

4. Obstructions to closing, read on turns: the shift of a turn is the Burgers vector of the singularities on one side, in the order of first visit of a cutting tree; a pair of valences three and five, or two vertices of equal valence exchanged, joined by a staircase of arcs between two turns, forces one of them to be shifted; a separatrix that winds once round and stops beside its own vertex does the same to a single turn; a checkerboard rule gives the parity of the shift between two turns, and on a closed sphere the shift of every turn has the parity of its length, at all balanced lengths; *combs* write every certificate of a conflict on the body as local dislocations (Section 8). Holonomy, Burgers vectors, dislocations as pairs of disclinations, the parity of cycles of quadrangulations and the Fox calculus are classical; their statements in terms of turns, staircases and combs are ours.

5. A lower bound on the number of turns of a family that every quantization within given bounds leaves shifted, by disjoint infeasible sets, with chains of dislocations as a corollary, computed exactly or within certified intervals on a human body and compared with the engine (Section 9). The bound is the packing bound of a minimum cover of irreducible infeasible subsystems, a known notion; its use for helices is new. Choosing the turns to close is NP-hard: known for maximum feasible subsystems in general, shown here for the restricted instances of quantization (bidirected balances, bounds 0 and 1).

6. An exact look at whole lengths on the same body: the turns, not the rounding, decide what closes (Appendix F).

7. Machine-checked proofs, in Lean 4 [17] with Mathlib [54], of the algebraic and combinatorial cores of the proofs, with the topological facts as explicit hypotheses (Section 10).

The results are preliminary: single runs except where three budgets are stated, one implementation, and heuristics where an exact formulation is still missing (Section 11). Statements are *proved* when they come with a proof, *measured* when they come from exact arithmetic or solver output on the systems openclare exports, and *conjectured* otherwise; we say which.

2 Background and related work

Counting singularities. On a closed surface of genus g, the valences $k(v)$ of a quad mesh satisfy $\sum_v (4-k(v)) = 4\chi = 8(1-g)$. With only valences 3 and 5, $n_3 - n_5 = 8(1-g)$: a sphere or a closed human body needs at least eight vertices of valence three. This bound says how many singular vertices are needed, not where they go; a hand-made human base mesh has hundreds.

Quad meshes as quartic differentials. Lei et al. [40] show that a closed quad mesh induces a Riemann surface and a meromorphic quartic differential $\omega = (dz)^4$ whose zeros and poles are the singular vertices, a vertex of valence k being of order $k - 4$; conversely a quartic differential with finite trajectories induces a quad mesh. The divisor of singularities then satisfies the Abel–Jacobi condition, and a divisor satisfying it is the divisor of a meromorphic quartic differential. Zheng et al. [61] optimize the singularity configuration by integer programming to satisfy the condition. Two points matter here. First, the condition is stated with holomorphic one-forms, which a surface of genus zero does not have; our reading is that in genus zero it adds nothing to the degree condition (this is an inference, not a statement of the papers). Second, the paper lists as an open problem the guarantee that all trajectories are *finite*. A trajectory that is not finite is precisely an edge loop that does not close: the helices of this note.

Closed trajectories. A quadratic differential all of whose non-critical horizontal trajectories are closed is a Jenkins–Strebel differential [53]; the complement of its critical graph is a union of flat cylinders (ring domains), each with a circumference and a height; we call *twist* the shift with which a cylinder is glued back after a cut along its core curve. In a quad mesh, a ring of quads is such a cylinder; a helix is a cylinder whose gluing is shifted by a whole number of rows. Section 3 makes this precise at the level of a T-mesh.

Quad meshes as flat metrics. Declaring every quad a unit square turns a quad mesh into a flat metric with cone points at its irregular vertices. Chen et al. [15] characterize the metrics that arise this way by four conditions: Gauss–Bonnet, rotational holonomy in multiples of a quarter-turn, alignment with the border, and a *finite geodesic lamination*: the straight lines of the field are finite loops. On a surface of genus zero the holonomy condition follows from Gauss–Bonnet (Lemma 3.9 of the arXiv version of [15]). The translational part of the holonomy is not discussed, and the finiteness condition is assumed, not constructed. Lei et al. [41] show that the singularity configurations of quad meshes are a strict subset of those of cross fields, and explain in this way why no quad mesh of the torus has exactly one vertex of valence three and one of valence five.

Dislocations. Izmestiev et al. [29] give geometric proofs of such non-existence results, among them the torus with one vertex of valence three and one of valence five, proved combinatorially by Barnette, Jucovič and Trenkler [6]. Their tool is the holonomy of the developing map of the flat metric, a rigid motion per loop. When the rotation of a loop is the identity, its translation is the *Burgers vector* of crystallography; two opposite cone points at distinct positions always have a non-zero one, which is the dislocation a pair of valences three and five makes in the square grid. Section 8 uses this tool for the shifts of turns.

Helices in quad meshes. Bommes et al. [8] define a *q-helix* as a dual parametric line of a quad mesh with an integer *pitch* h and *turn length* τ: walking τ steps along it is equivalent to h side-steps. They detect minimal q-helices ($h \leq \tau$, no sub-helix of smaller pitch) by the self-intersections of orthogonal dual lines, and remove them greedily with *grid-preserving operators*: closed dual paths of shift and collapse steps, found by a graph search, that keep the mesh all-quad and its singularities in place. Tangential smoothing then repairs the distortion the shifts introduce. Their experiments remove most helices (from 17–52 down to 1–23 per model) and coarsen the base complex accordingly. They note that choosing the pairs of singularities to align directly in the parametrization, as Myles et al. do for T-meshes [47], is often complicated or even ambiguous to automate. The approach works on a finished mesh; it does not touch the parametrization or its quantization. The closest prior work after it is that of Razafindrazaka and Polthier [51], who start from a finished quad mesh and replace its base complex by an optimal one, found as a perfect matching on a graph over its singular vertices; they note that

the operators of [8] leave some spirals. Like those operators, this works after meshing, not in the quantization.

T-mesh quantization. Mixed-integer quadrangulation [7] and integer-grid maps [9] decide the integer structure of a quad mesh in their quantization. Campen et al. quantize via the integer lengths of the arcs of a T-mesh traced in a seamless parametrization [13]. Lyon et al. solve a constrained version as an integer linear program, bounding the directional deviation of the separatrices of the resulting layout [45], and require, as openclare does, that the quantization keep singularities apart. Heistermann et al. cast quantization as a minimum-deviation flow on a bi-directed graph and give a fast approximate solver and an exact refinement [26]; openclare follows their formulation. To our knowledge none of these constrains the helices: the lengths are chosen for consistency, validity and closeness to the parametrization, and the loops close or spiral as a side effect.

Periods round a tube. QuadCover [32] lifts the cross field to a branched covering where it becomes a vector field, and integrates it there. The stripe parameterization of tubular surfaces [33], built on it, measures the jump ("gap") of the parameter across each cut path, which is constant along the path, and rounds it to the nearest whole number; the grid lines are continuous exactly when every gap is whole. The gap of the coordinate across a loop round a tube is what we call the shift of a turn: rounded to zero it gives rings, rounded to a non-zero whole number a helix of that pitch. That paper treats stripes spiralling round the tube as a legitimate outcome, and notes that rounding the gaps for least distortion is a hard combinatorial problem that it does not solve exactly.

Layouts traced from the field. Myles et al. [48] trace the field into a partition into quadrilateral patches and adjust the layout until its lengths are consistent; Pietroni et al. [50] select traced separatrices by a binary linear program; Razafindrazaka et al. [52] obtain a layout as a minimum-weight perfect matching on a graph of the singularities. Campen surveys these methods [11]. Except for the parametrization of [48], they build coarse layouts; whether a dense mesh made from the layout spirals is, as far as we read, not among their criteria.

Lower bounds. We found no result that bounds from below the number of helices of a quad mesh, or the number of its loops that do not close, for a given singularity configuration. The nearest results are existence and non-existence theorems: Strebel's existence of Jenkins–Strebel differentials with prescribed cylinders on a fixed Riemann surface [53], the non-existence results above [29, 41], and the finiteness condition of [15], which the open problem of [40] asks to guarantee. Our bound is a minimum cover of irreducible infeasible subsystems, a classical notion of linear feasibility [3, 49], applied to the shifts of turns (Section 9).

Loops by construction. Dual loops meshing [12] builds a layout from loops traced on the surface, which close by construction; it produces coarse layouts rather than dense field-aligned meshes. Lei et al. [39] build quad meshes from a Strebel differential with prescribed cylinders, so that every trajectory closes, and skeleton-guided layouts [56] close the loops round limbs by design. Both produce coarse layouts, and neither starts from a given cross field and singularity configuration.

Measuring loops. QuadGPT [43] counts *quad rings* (closed loops of faces) and *quad lines* (open strips) and uses the share of faces in rings and the mean length of lines as reward signals for preference optimization; it does not report them as evaluation metrics, nor does it tell spirals from open lines. SQuadGen [38] scores the face loops and edge loops of a layout by their self-intersections and their rotation index (the total curvature of the loop, projected on a best-fit

plane, over 2π), and reports the area share of simple loops; it does not tell helices from open loops either.

3 Turns of a T-mesh

Setting. A T-mesh is a partition of the surface into rectangles, the *patches*, whose sides are made of *arcs*; nodes where a trace stops on another are T-junctions, and the singular vertices of the field are nodes. Each patch P has local coordinates with its first corner at the origin and its sides along the axes. A quantization assigns every arc a a length $\ell_a \in \mathbb{Z}_{\geq 0}$ such that opposite sides of every patch have equal length; the quad mesh is then read off as the integer grid of each patch.

Definition 1 (Turn)**.** *A* turn γ *is a straight walk across patches, from an offset of its first side that meets no node: it enters a patch through a side, leaves through the opposite side at the same distance from the corresponding corner, enters the neighbouring patch across the arc it reaches, and so on, until it comes back into its first patch through its first side. The rotation of the composed crossings is then the identity, since the walk re-enters its first side inward.*

Crossing from patch P to patch Q across an arc composes a quarter-turn rotation of the local frame with a translation whose components are signed sums of arc lengths along the rims of P and Q: the position of a rim point is the sum of the lengths of the arcs between it and the first corner.

Proposition 1 (The shift of a turn is linear)**.** *For a turn γ there are integer coefficients c_a, independent of the lengths, such that after a full turn a point of the first side has moved across the walking direction by*

$$s_\gamma(\ell) = \sum_a c_a \, \ell_a.$$

For integer lengths ℓ, let o be a half-integer offset of the first side whose walk at the lengths ℓ crosses the same patches as γ (we call the set of such offsets the corridor *of γ at ℓ). Then the dual line of the quad mesh at o comes back at $o + s_\gamma(\ell)$: it closes after one round if and only if $s_\gamma(\ell) = 0$. If moreover $o + j\, s_\gamma(\ell)$ lies in the corridor for $j = 1, \ldots, t$, it winds t times round as a helix of pitch $|s_\gamma(\ell)|$ in the sense of [8]. If the corridor holds no half-integer, no strip of the mesh follows γ.*

Sketch. Each crossing is a map $x \mapsto R_k x + t$ with R_k a rotation by k quarter-turns and t a vector whose entries are sums of $\pm\ell_a$. The composition along γ has trivial rotation, hence is a translation $T_\gamma(\ell)$, linear in ℓ with integer coefficients. Its component along the walking direction does not move a point of the first side off that side; its component across is $s_\gamma(\ell)$. The walk crosses no singularity (singularities are nodes, never inside a patch), so the quad mesh is a regular grid along it, and a dual line at distance o from the first corner, as long as it crosses the same patches as γ, comes back at distance $o + s_\gamma(\ell)$. The corridor is cut out by two inequalities per crossing (the walk must cross each arc strictly between its ends), so it may contain o but not $o + s_\gamma(\ell)$: the line then comes back $s_\gamma(\ell)$ rows beside itself, but its next round follows other patches, and on a closed surface it may close after visiting them (Proposition 2). □

The shift form is thus a statement about a fixed sequence of crossed patches. Closing constraints $s_\gamma(\ell) = 0$ describe the strips of the mesh only together with the corridor inequalities of γ, which the engine of Section 4 does not impose (Appendices F and G measure what this costs). Explicitly, let γ cross the arcs $a_1, \ldots, a_m$ in turn, and let $p_j(o, \ell)$ be the distance from the first corner of the side holding a_j at which the walk from offset o crosses that side. Each crossing map is an isometry whose translation is an integer form in ℓ, so $p_j(o, \ell) = \epsilon_j \, o + q_j(\ell)$

with $\epsilon_j = \pm 1$ and q_j an integer linear form; let $e_j(\ell)$, a sum of arc lengths, be the distance of the start of a_j along the same side. The corridor is the interval

$$\{\, o \;:\; e_j(\ell) < \epsilon_j\, o + q_j(\ell) < e_j(\ell) + \ell_{a_j}, \quad j = 1, \ldots, m \,\},$$

and at whole lengths, for a half-integer o, each strict inequality is the same as one at distance $\frac{1}{2}$ from the ends of the arc: $e_j(\ell) + \frac{1}{2} \le \epsilon_j\, o + q_j(\ell) \le e_j(\ell) + \ell_{a_j} - \frac{1}{2}$. These are the inequalities that Appendix G imposes.

The shift is the across component of the translational holonomy of the flat metric along the turn (Section 8.1), which is linear in the lengths as any period of a flat cone metric [55, 57], and for a ring of quads it is the twist of a cylinder. What is specific here is its expression in the arc lengths of a T-mesh, as a constraint of the quantization.

The component of $T_\gamma(\ell)$ along the walking direction is the *length* $\alpha_\gamma(\ell)$ of the turn, the distance walked before coming back to the first side; it is also a linear form in ℓ with integer coefficients, and for whole lengths it is the number of quads of the strip in one turn.

In openclare (`quantization/tubes.rs`) the shift is built once per turn as an expression in the arc lengths, by composing the rim expressions of the crossed patches; evaluating it with the real lengths of the parametrization gives the *real shift*, the twist the field itself would give the cylinder.

Definition 2 (Strip)**.** *For whole lengths ℓ balancing every patch, with every patch side positive and no two singular vertices at the same point, the quad mesh is, in each patch, the integer grid of its sides. A* strip *is a maximal sequence of quads, each sharing with the next the edge opposite the one it shares with the previous.*

Proposition 2 (Completeness at given lengths)**.** *Let ℓ be whole lengths balancing every patch, with every patch side positive and no two singular vertices at the same point. Every strip of the quad mesh that crosses a side σ of a patch P at the quad edge $[k, k+1]$ of σ is the walk of Definition 1 taken* with the lengths ℓ *from σ at offset $k + \frac{1}{2}$. In particular, on a closed surface, walking from every half-integer of every side finds every strip of the quad mesh: the strips are the cycles of the walk on the half-integer stations, each oriented strip one cycle; by Proposition 1, with γ the turn the strip follows from a station, the strip closes after one round exactly when $s_\gamma(\ell) = 0$. With borders, the strips that reach a border are open paths of the walk.*

Proof. Inside P, the quads of the strip are the unit squares between the grid lines at k and $k+1$ across σ; their dual line is the straight line at $k + \frac{1}{2}$, which reaches the opposite side at the same distance from the corresponding corner, which is the move of the walk. At the arc where it leaves P, the crossing map to the neighbouring patch is a quarter-turn followed by a translation whose components are sums of $\pm\ell_a$, hence integers: it maps the grid of P onto the grid of the neighbour, and the midpoint of a quad edge onto the midpoint of a quad edge. Every node lies at a whole distance along the sides, so a walk at a half-integer offset never meets a node, and the arc it crosses is determined. A single arc may have length zero (two T-junctions, or a T-junction and a corner, at the same point): it holds no half-integer, so no walk crosses it, and nothing in the argument changes; only the sides of the patches need to be positive. The strip and the walk thus continue together, patch after patch. A strip only meets singular vertices at its sides, never inside, since they are corners of patches. Positive sides make each patch a grid with at least one quad in each direction, so that every station lies on a quad edge of the mesh. □

The sampled walks of Section 4 start at fixed shares of the real side lengths. Proposition 2 says where the helices of a given mesh are *all* found: at the half-integers of the whole lengths. The engine uses this to generate closing constraints (Section 4).

4 Closing turns in quantization

This section describes the engine at commit `3b890ecc8080`, in `crates/openclare/src/quantization/`. Some experiments of later sections used earlier versions; each says which.

The quantizer of openclare minimizes a separable convex cost of the integer lengths, in the minimum-deviation-flow formulation of [26], subject to the balance of each patch, by cancelling improving walks in a strip network (a simpler solver than their matching-based one). Only the structure is theirs: their objective is a sum of absolute deviations, ours is quadratic. The cost of arc a is $w_a(\ell_a - \lambda r_a)^2$ with $w_a = 1/(\lambda r_a + 1)$, plus a term that draws a vertex onto each fold of the surface the arc crosses, and a penalty for going below a lower bound (the mean absolute deviation, an L^1 measure, is used afterwards, as a cap in the selection of candidates); the real lengths r_a carry a fixed per-arc perturbation of at most 0.5%, so that arcs of equal length do not all change at the same scale. The scale λ is searched for the number of quads asked for, and arcs whose zero length would break the structure of the mesh are held at length at least one. Closing turns is a side constraint that is not a flow constraint. We handle it in four steps.

Finding turns. From points at given shares along each side of each patch, we walk straight across patches and keep the walks that return. A turn is *held* if its real shift is above 0.05 and its lean below a bound, and if its length is below a multiple of the side of the T-mesh (the side of a square of its area). Two turns with the same shift expression up to sign are one ring and held once. Eight *ways* are tried: starting shares $\{0.5\}$ or $\{0.2, 0.5, 0.8\}$, lean bounds 0.05 or 0.12, length bounds 2 or 6 sides; a way that holds the same turns as an earlier one is skipped.

Least-change projection. The target lengths are changed as little as possible, in the weighted least squares sense, for every held turn to have zero shift and every patch to stay balanced:

$$\min_x \sum_a w_a \,(x_a - r_a)^2 \quad \text{s.t.} \quad Ax = 0, \quad s_\gamma(x) = 0 \;\; \forall \gamma \in H,$$

solved with a sparse Cholesky factorization of the balances, computed once per quantization, and a small dense Schur complement for the turns. Arcs the solution would make negative are held at zero and the system solved again (an active set). If the system cannot be met, the held turn left with the largest shift is let go and the projection repeated.

Integer search and repair. The integer search of the quantizer runs on the projected targets. Rounding leaves some turns shifted by one or a few rows. Repair walks are then taken one at a time: a walk is guided by a bias along the arcs of the shifted turns, and accepted if it reduces the total shift, costs little in deviation, keeps the structure intact and the number of quads within tolerance. A second, *aimed* repair treats each held turn still shifted on its own: the bias is put on its arcs alone, and a walk is accepted if it shifts that turn less and the others no more. On the pieces that are closed spheres, walks that cannot close a turn by the parity of Corollary 1(c) are turned down without being judged, and a turn whose shift is a non-negative form already at its least (Remark 4) is not aimed at; the engine also checks, in its debug builds, the integer coefficients of the shifts, the handshake of Lemma 5 and the parity of Proposition 6.

Rounds of constraint generation. Proposition 2 says where the strips of a given mesh are *all* found: at the half-integers of the whole lengths. After a way whose lengths are admissible (within the budget and the deviation cap below), up to three rounds follow, a lazy generation of constraints (the equations are added as the current solution violates them; they are equalities on the shifts, not integrality cuts): every strip of the whole lengths is walked (up to 6 times the side of the mesh), those that come back shifted although the real lengths bring them back within 0.12 of their length are held as well, and the projection and the integer search are run

again from the lengths of the round before. In these rounds the projection does not use the active set and the repair is not aimed. Each round gives one more candidate.

Selection and safeguards. For every candidate, the quad mesh connectivity is built and the share of edges on spiral loops measured (Section 5). A candidate is kept if it spirals less than the plain search (without closing), its number of quads is within the tolerance (or no further from the budget than that of the plain search), its mean deviation from the real lengths is at most $\max(1.5\,d_0,\ d_0 + 0.2)$ with d_0 that of the plain search, and it leaves the vertices no further from the folds of the surface than the plain search did, by more than chance would. The three with the fewest spirals are kept, in that order, then the plain lengths. The ways are independent and run in parallel. After meshing, if the first candidate makes a face fold over the surface, the others are meshed in turn until one does not, and the mesh with the fewest folds, then the fewest spirals, is kept.

Remark 1. *A turn round a part of the shape whose real shift is far from zero is not a rounding artefact: the field winds round the part. We only hold turns whose real shift is a small share of their length, the* lean*; a turn round a group of singularities, by contrast, may legitimately come back shifted.*

5 Measuring loops

An edge loop is followed straight through every regular vertex (valence four, interior), each edge belonging to one loop. A loop is *closed* if it comes back to its start; *border* if it runs from border to border; *open* if it stops at an irregular vertex or a non-quad face; and *spiral*, whatever its ends, if a strip of quads leaving it across meets it again, the same way round, within $k_{\max} = 4$ quads. The last test catches helices of pitch up to four, including bundles of interleaved helices [8], while a ring is met again only round a whole torus or over a cap the other way round. We report the share of edges on each kind of loop, edges of borders left out.

6 Experiments

All numbers come from the openclare bench (repository `marcgardent/openclare`), measured on the output meshes against their sources. The engine is deterministic: one run per model and budget. Models: twelve organic and mechanical models for development, and a corpus of 33 models for the full bench, among them two human bodies from Blender's Essentials library (CC0) whose base meshes, modelled by hand, serve as references. Appendix A gives the engine version and the source of every figure.

Hardware and solvers. Every time in this note was measured on one machine: an Intel Core i7-7700K (4 cores, 8 threads, 4.2 GHz, 2017) with 62 GB of memory. The machine was shared with other jobs (other experiments and builds running at the same time) for most measurements, and the load was not recorded; times are wall-clock times and give orders of magnitude only, and outcomes that depend on a time limit depend on this hardware and load. Linear and mixed-integer programs were solved with HiGHS 1.8.0 [28] through SciPy 1.16.3 [58] (`scipy.optimize.linprog` and `milp`), with SciPy's default options (threads included) except the time limits stated.

Spirals on the twelve development models. The share of edges on spiral loops is chaotic on a single model: a change of one per cent of the budget moves it as much as a real change of the method. We therefore judge the engine by its mean over the twelve models at three budgets, 2,970, 3,000 and 3,030 quads (36 runs). At the cited commit the engine leaves 4.65% of the

edges on spiral loops (2.89%, 4.97% and 6.09% at the three budgets; 102 spiral loops in the 36 meshes, all valid), with 36.7% of the edges on closed loops, a mean distance to the source of 0.377‰ of the diagonal and a mean corner deviation of 4.90 degrees. The development steps were judged, at the time, on single runs at 3,000 quads (Table 1): from 16.7% of spiral edges without closing to 6.0%. Over those steps the Hausdorff distance stays within about 2% and the mean corner deviation within about 6%; the last step trades a little of both for a third fewer spiral edges.

	spiral edges	Hausdorff (‰)	corner deviation (deg)
no closing	16.7%	8.77	4.75
closing, one way kept by spirals	9.7%	8.57	4.85
+ leaning ways, aimed repair	9.1%	8.58	4.91
+ longer turns, deviation cap	6.0%	8.68	5.01

Table 1: The development of closing: mean over the twelve models at 3,000 quads, one run per model, engines of commits `7dc1727`, `c86740f` and `cf4a82c`. Each line adds to the one above only roughly: between the second and third lines the baseline moved (commit `ab8d89c`, 10.1% to 9.5%). The constraint rounds, the active set and the floor of the cap came later; the engine at the cited commit gives 4.65% over three budgets (text).

Against other remeshers. The last full bench committed (`bench/results/results.json` at commit `83f0967`, 3,000 quads; one run per model for openclare and AutoRemesher, the median of five seeds for QuadriFlow [27], run through Blender) was made with an engine that differs from the cited one by a few changes; on the twelve models at three budgets these move the mean from 4.54% to 4.65%. The tools do not mesh the same models: QuadriFlow produces a mesh for 22 of the 33 models, AutoRemesher [21] for 18. On the 7 models that all three mesh validly in every run, the mean share of spiral edges is 1.8% for openclare, 38.8% for QuadriFlow and 21.7% for AutoRemesher; on the 15 models where all three produce a mesh, valid or not, 4.2%, 27.2% and 27.3%. Without closing (commit `4e9620c`) openclare had 16.6% and 15.5% on the same subsets.

Against the hand-made mesh. The male body of the Essentials library has a base mesh of 10,590 quads modelled by hand, with 2.5% of spiral edges in 3 spiral loops and 31.6% of its edges on closed loops. Remeshed by openclare at that count with the first version of closing, whose only lean bound was 0.05, it had 12.1% (one leg was a single helix from hip to ankle, its turns leaning more than that bound allowed; commit `ab8d89c`). Single runs of the engine from commit `b0d37fa` to `8fd1932`, with the closing of Section 4, give 1.6% of spiral edges, in 9 spiral loops: below the hand-made mesh, but by a few loops only. (The later changes up to the cited commit, a finer work mesh at thin walls, were not measured on this body; earlier versions gave from 1.1% to 2.4%.) One such run has 33.2% of its edges on closed loops, a mean distance to the source of 0.19‰ of the diagonal and a mean corner deviation of 4.1 degrees, against 0.17‰ and 9.2 degrees for the hand-made mesh. These fidelity figures compare meshes made for different ends: the hand-made mesh is a base mesh meant for further modelling, and its distance to the source and its corner angles are choices of its modeller, not defects.

The quad sphere. Not a result of closing but of the field (commit `18bc18c`). An icosphere remeshed at 3,000 quads had 32 irregular vertices, the pattern of its own facets, because the field was pulled towards curvature directions that a sphere does not have; at 1,000 quads it had the eight poles but no structure (a base complex of 1,010 patches) and 87% of spiral edges. With

directions not trusted where the two principal curvatures are within 10% of one another, both give the layout of a cube: eight poles of valence three, six patches, no spiral.

7 An exact look at the closing system

To see how far the projection of Section 4 is from what the constraints allow, an experimental hook of openclare (branch `rd-exact`, on an engine before the constraint rounds and the floor of the deviation cap of Section 4: commit `3b3122d` and later are not in it) exports, for each way of finding turns, the system it works on: the real lengths of the 2,069 arcs of the T-mesh of the male body at 10,590 quads, the 1,552 balance equations of its patches, every turn found with its shift, and the lower bounds the integer search placed on arcs to keep singularities apart (28 to 36 arcs held at length one). These bounds are the engine's choice of *which* arc of a chain between two singular vertices carries their separation: validity asks a chain to keep a positive total length, not a given arc. Every statement below that uses them is relative to them; we call them the *engine's bounds of separation*, and P_{eng} the balanced lengths that meet them. P_{eng} is neither a subset nor a superset of the valid quantizations: it excludes valid lengths that put the separation on another arc of the chain, and it allows lengths that bring other arcs to zero and break the mesh. We solve it with HiGHS through SciPy, in real numbers. The ways are numbered as exported: way $m + 2j + 4i$ starts from the shares $\{0.5\}$ ($m = 0$) or $\{0.2, 0.5, 0.8\}$ ($m = 1$), with the lean bound 0.05 ($j = 0$) or 0.12 ($j = 1$) and the length bound 2 ($i = 0$) or 6 ($i = 1$) sides. Way 2 is the one the engine keeps for this body.

Where the conflict lies (measured). All held turns can be closed together if the lengths are only asked to be non-negative. With the bounds of separation, they cannot. In four ways (0, 1, 2 and 4) one held turn cannot even be closed on its own: the same turn round the left wrist, of length 7.0, whose shift is a sign-definite form in two arc lengths (Section 8, Remark 4). Giving up the least total amount of the bounds needed to close all held turns, a linear program, shows that in ways 0, 2 and 4 the bound of a *single* arc stands in the way, the arc of that wrist turn, 0.08 quad edges long; Section 9 shows that in ways 0, 2, 4 and 6 letting go of a single turn is enough, and in the others of four to nine (Table 6). The conflict between closing loops and keeping singularities apart is thus real, but local.

How far the heuristic is. With every bound kept, the largest set of held turns that can be closed together is found by a mixed-integer program with one binary per turn (Table 2). The projection of the engine of the export keeps far fewer: it removed, one at a time, the turn left most shifted by a least-squares projection whose negative lengths were clamped to zero, and a single clamped arc could make it drop dozens of turns that conflict with nothing. The engine at the cited commit holds such arcs at zero instead (the active set of Section 4), the variant tested in Table 3.

In whole numbers, the same mixed-integer program did not finish within fifteen minutes on the two systems tried. Appendix F shows that a similar failure, on a larger set of rings, came from the way the problem was posed to the solver and not from an absence of whole solutions.

From reals to whole numbers. We then fed the engine better targets and measured what survives its integer search (Table 3). Aiming at the lengths of the linear program closes fewer loops in the end: the weighted L^1 objective puts the changes on few arcs, the deviation from the real lengths grows, and the variants with the fewest spirals are refused by the deviation cap. A projection that holds lengths at a floor instead of clamping them (an active-set method on the same Cholesky solver) keeps every held turn closed in real numbers; yet after the integer search 117 of 169 are shifted again, and the mesh spirals more. With the clamped projection,

way	held	kept by the projection	closable together
0	48	42	47
1	78	46	73
2	52	46	51
3	86	47	82
4	84	49	83
5	160	48	154
6	88	53	87
7	169	49	160

Table 2: Male body at 10,590 quads, real numbers, engine of the export. "Kept by the projection": by the clamped projection of that engine; "closable together": the largest set of held turns closable with every bound of separation kept.

too, about 40% of the turns it kept are shifted after the integer search. The bottleneck is not the projection but the rounding: the integer search follows the targets but ignores the shifts.

targets of the integer search	turns closed in reals	still shifted after	spiral edges (mesh)
projection, clamped	about 50	10–23 of them	1.1%
linear program, L^1	up to 160	—	11.9%
active set, floor 0	all held (up to 169)	117 of 169	3.3%
active set, floor half the real length	36–72	—	9.8%

Table 3: Male body at 10,590 quads, engine of the export, one run each; the mesh keeps the way with the fewest spirals, so its share is that of a single way. The engine at the cited commit uses the active set with floor 0 for the turns of each way, together with the constraint rounds and the floor of the deviation cap, which this table does not include.

Whole lengths. Appendix F looks at the same body in whole numbers. The balances are the incidence matrix of a bidirected graph, whose free arcs parametrize the lattice of whole balanced lengths (Proposition 18): balances alone never obstruct whole lengths. Solver failures in whole numbers came from implicit zeros, arcs that every real solution sets to zero, not from an absence of whole solutions. The obstruction that remains is the size: closing the 218 ways round of the mesh of way 2 that lean like rings needs a mean deviation from the targets of at least 0.817, far above the engine's cap; a subset of 190 closes at 0.399 in real numbers and needs at least 0.536 in whole numbers (proven by the solver), and the whole lengths found close all 190 but make a mesh that spirals more than the engine's, because the strips move with the lengths. On this body the choice of the turns, not the rounding, decides how much the mesh spirals.

8 When rings cannot all close

Section 7 found that closing all held turns conflicts with keeping singularities apart, and that the conflict comes down to one or a few arcs. This section gives a mechanism for such conflicts, with proofs. The tool is the holonomy of the flat metric, as used by Izmestiev et al. [29]; its ingredients are classical, and we know of no earlier statement of the propositions below in terms of helices. Appendix B tests them on the male body, and Section 8.2 describes its conflicts.

8.1 Turns and singularities

Flat metric and holonomy. In Sections 8.1 to 8.4 and Appendices B to D the surface Σ is closed, without border, and *genus zero* means a closed sphere. Several statements fail with borders or handles; Section 8.5 gives their general form. Lengths $\ell > 0$ balancing every patch make each patch a Euclidean rectangle. Glued along their arcs, whose lengths agree on both sides, the rectangles give the surface Σ a flat metric with cone points at the singular vertices: a vertex of valence k has angle $k\pi/2$ and curvature $(4-k)\pi/2$; a T-junction has angle 2π and is not a cone point. Let Σ° be Σ without its singular vertices, x_0 a point inside a patch, and R the quarter-turn in the positive sense of the orientation of Σ. Developing a loop γ of Σ° based at x_0 patch after patch, as in the proof of Proposition 1, gives a motion $\hat{h}(\gamma)\colon x \mapsto R^j x + t$ of the coordinates of the first patch, its *holonomy*: the chart φ of x_0, continued along γ, comes back as $\hat{h}(\gamma) \circ \varphi$. It depends only on the homotopy class of γ in Σ°, and $\hat{h}$ is a homomorphism of the fundamental group into the group of such motions [29], $\hat{h}(\alpha\beta) = \hat{h}(\alpha) \circ \hat{h}(\beta)$. A small loop round a singular vertex c of valence k, run with c on its right and reached along a path that develops it at the point p, has holonomy $x \mapsto R^{4-k}(x-p)+p$: turning clockwise round a cone point of angle $k\pi/2$ rotates the chart by $-k\pi/2$, which is $(4-k)\pi/2$ modulo a full turn. When the rotation of $\hat{h}(\gamma)$ is the identity, its translation is the *Burgers vector* of γ.

For a turn γ, let $\bar{\gamma}$ be the closed curve that follows the walk and, back in its first patch, returns to its starting point along a segment parallel to the first side, just inside the patch. Let e_1 be the walking direction and e_2 the direction along the first side. By the proof of Proposition 1, $\hat{h}(\bar{\gamma})$ is a translation and

$$s_\gamma(\ell) = \langle e_2,\ \hat{h}(\bar{\gamma}) \rangle.$$

Shifts, developed positions and Burgers vectors are linear forms in ℓ with integer coefficients, as long as the walks and paths cross the same patches. An identity between them that holds on an open set of balanced positive lengths therefore holds for all balanced lengths, and for whole lengths the developed positions of the nodes are points of $\mathbb{Z}^2$.

Lemma 1 (Gauss–Bonnet for turns)**.** *Let γ be a turn whose curve $\bar{\gamma}$ is simple. If $\bar{\gamma}$ bounds a disc $D \subset \Sigma$, the singular vertices inside D satisfy $\sum_{c\in D}(4-k(c)) = 4$. If the curves of two turns are disjoint and bound an annulus A, then $\sum_{c\in A}(4-k(c)) = 0$.*

Proof. This is the discrete Gauss–Bonnet count, standard in quad meshing [10], applied to the curve of a turn. For a region of a flat surface with cone points, bounded by straight segments, Gauss–Bonnet reads $\sum_c \kappa_c + \sum_i \theta_i = 2\pi\chi$, where κ_c are the curvatures of the cone points inside and θ_i the turning angles at the corners of the border. The curve $\bar{\gamma}$ is straight except at the two ends of its closing segment, where it turns by $+\pi/2$ and $-\pi/2$ (or not at all if the shift is zero); the turning angles sum to zero on each curve, seen from either side. With $\kappa_c = (4-k(c))\pi/2$, $\chi(D) = 1$ and $\chi(A) = 0$ the two statements follow. □

On a closed surface of genus zero, such as a body, every simple closed curve bounds a disc on each side. Every turn with a simple curve thus splits the singular vertices into two groups of total index four each, consistently with the total $8(1-g) = 8$ of Section 2: a fingertip beyond a ring holds the equivalent of four vertices of valence three. A loop round a group of singularities whose indices do not add up to four is never a turn, whatever the lengths. A side holding at most one singular vertex has total index 0 or $4-k \neq 4$, so each side of a turn holds at least two. None of this holds with borders or handles: a side holding b border components without corners and h handles has total index $4(1-b-2h)$, and may hold no singular vertex at all (Lemma 4).

Curves of turns need not be simple, and on the male body most of those held for closing are not (Appendix B). The first case beyond simple curves is still determined.

Lemma 2 (One self-crossing)**.** *Let the surface be closed of genus zero, and let the curve $\bar{\gamma}$ of a turn cross itself exactly once, transversally, with no singular vertex on it. It cuts the surface into*

three discs: two lobes*, each bounded by one of the two loops of $\bar\gamma$, and a third region bounded by both. If the crossing lies on the walk, the totals of $4-k$ over the singular vertices of the lobes and the third region are* 3, 3 *and* 2*; if it lies on the closing segment, they are* 2, 2 *and* 4*.*

Proof. A walk follows the two directions of the field, so it crosses itself at a right angle. Each loop leaves the crossing point X along one passage and comes back along the other, so its two rays at X are a quarter-turn apart. A loop is simple; it bounds on one side a disc containing the quadrant between its two rays, and the other loop, which meets it only at X, lies in the closure of the other side. That disc is a lobe, with a corner of angle $\pi/2$ at X; the two lobes take opposite quadrants, and the third region has the other two. The curve is a connected graph on the sphere with one vertex and two edges, so its three faces are discs. By Gauss–Bonnet as in Lemma 1, a disc whose border turns by the angles θ_i at its corners holds singular vertices of total index $4-\sum_i \theta_i/(\pi/2)$. A corner of angle $\pi/2$ turns by $\pi/2$, which gives 3, 3 and 2 before the corners of the closing segment, at which the curve turns by $+\pi/2$ and $-\pi/2$ in succession. If the crossing lies on the walk, both of these corners are on one loop, every region sees both, and they cancel. Otherwise let the walk start at A along e_1 and come back at Z, and let it cross the closing segment between them. It crosses walking along $-e_1$: walking along e_1, it would enter the first patch through its first side there, and the turn would have ended. The loop through A then turns the same way at A as at X, and so does the loop through Z at Z and at X (a direct check of the four directions). Each lobe lies on the side of its corner of angle $\pi/2$ at X, so it sees a second convex corner and holds $4-2=2$; the third region holds the rest of the total 8, that is 4. □

With borders or handles the regions need not be discs, and the totals follow region by region from Lemma 4: turns with one self-crossing split the singular vertices as $(3,3,-6)$ on Suzanne, $(3,3,-2)$ and $(3,2,-1)$ on Jerry, and, on the genus-one Bob, into two regions as $(3,-3)$ and $(2,-2)$ (measured).

Proposition 3 (The shift is set by the singularities)**.** *Let γ, γ_1, γ_2 be turns with simple curves.*

(a) Let $\bar\gamma$ bound a disc D holding the singular vertices $c_1,\dots,c_m$, and let x_0 be a point of $\bar\gamma$, the start of the walk for instance. Let $T\subset D$ be a tree whose vertices include $c_1,\dots,c_m$ and which meets $\bar\gamma$ only at x_0. Cut along T, the disc D becomes a disc Ω without cone point, which develops into the plane, with the frame (e_1,e_2) at x_0. Number the c_i in the order in which a walk round T from x_0, keeping T on its right, first reaches them, and let p_i be the developed position of c_i at that first visit. Then $s_\gamma(\ell)=\pm\langle e_2,B_D\rangle$, with

$$B_D=\sum_{i=1}^{m} Q_i\,(I-R_i)\,p_i,\qquad R_i=R^{4-k(c_i)},\qquad Q_i=R_mR_{m-1}\cdots R_{i+1},$$

the translation of $\rho_m\circ\cdots\circ\rho_1$, where $\rho_i(x)=R_i(x-p_i)+p_i$; the sign is $+$ if D lies on the right of $\bar\gamma$, and $-$ otherwise.

(b) Let $\bar\gamma_1$ and $\bar\gamma_2$ be disjoint and bound an annulus A, round which the two turns go the same way, and let a path κ inside A from $\bar\gamma_1$ to $\bar\gamma_2$ carry the walking direction of γ_2 onto that of γ_1 (the turns are parallel*); measure both shifts along the direction e_2 carried by κ. Then $s_{\gamma_1}(\ell)-s_{\gamma_2}(\ell)=\langle e_2,B_A\rangle$, where B_A is the Burgers vector of the border of the disc A cut along κ, given by the same sum over the singular vertices inside A, developed inside the cut disc.*

(c) In particular, parallel turns bounding an annulus without singular vertex have the same shift for every ℓ: a helix is a property of a whole cylinder of parallel turns, not of one of them.

Proof. (a) Orient each edge f of T away from x_0. Its two sides on the border of Ω are glued by a motion G_f: a point of f that develops at u on the side the walk round T follows first develops at $G_f(u)$ on the other. Let f end at the vertex v, from which the edges $f_1, \dots, f_r$ leave in the order of the walk. Keeping T on its right, the walk turns clockwise round v, so a small arc round v from the first side of f to the second turns clockwise through the whole angle at v and crosses $f_1, \dots, f_r$ on its way. Continuing the chart of Ω along this arc composes it with $G_{f_1}^{-1}, \dots, G_{f_r}^{-1}$ at the crossings, and turning once round v composes it with the rotation $\rho_v(x) = R^{4-k(v)}(x - p_v) + p_v$, where p_v is the position of v at the first visit, the one the arc starts from. Comparing the two at the second side of f,

$$G_f = G_{f_r} \circ \cdots \circ G_{f_1} \circ \rho_v,$$

with ρ_v the identity if v is not singular. The same comparison at x_0, with the edges $f_1, \dots, f_r$ leaving x_0 and the arc round v replaced by $\bar\gamma$ run with D on its right, gives $\hat h(\bar\gamma) = G_{f_r} \circ \cdots \circ G_{f_1}$. Unfolding the recursion, $\hat h(\bar\gamma)$ is the composition of the ρ_{c_i}, each vertex before the vertices below it and the branches in the order of the walk: the order of first visit. Its rotation is $R^{\sum_i (4-k(c_i))} = R^4 = I$ by Lemma 1, and expanding the composition gives B_D; run with D on its left, $\bar\gamma$ has the inverse holonomy, the translation by $-B_D$. The shift is the component along e_2. When T is a star of disjoint paths from x_0, the order is the one in which the paths leave x_0, as the walk meets them. (b) Take x_0 on $\bar\gamma_1$ at the start of κ. The loop $\lambda = \bar\gamma_1 \, \kappa \, \bar\gamma_2^{-1} \kappa^{-1}$ is the border of the disc obtained by cutting A along κ. Developing along κ is a change of coordinates; conjugating the translation $\hat h(\bar\gamma_2)$ by it gives that translation read in the coordinates at x_0, where the walk of γ_2 runs along e_1 by hypothesis. Hence $\hat h(\lambda)$ is the translation by $\hat h(\bar\gamma_1) - \hat h(\bar\gamma_2)$, both read in the same coordinates, whose component along e_2 is the difference of the shifts, and part (a) applied to this disc, whose singular vertices have total index zero by Lemma 1, gives the sum. (c) Without singular vertex the sum is empty. □

The simplest configuration between two parallel turns is a pair of singular vertices of valences three and five, the defect that lets the number of edges round a limb change. It is a dislocation of the grid. That a pair of opposite disclinations is a dislocation, with a Burgers vector of length $2\sin(\Omega/2)$ times their separation for a rotation Ω, is classical in the theory of crystal defects [37, Eq. (5)]; on a quad mesh $\Omega = \pi/2$, and on a triangle mesh a pair of valences five and seven is the elementary dislocation of graphene [59]. What follows reads it on the shifts of two turns.

Proposition 4 (A dislocation between two turns)**.** *In the setting of Proposition 3(b), let the annulus A hold exactly two singular vertices, c_3 of valence three and c_5 of valence five, and let $d = p_5 - p_3 = d_1 e_1 + d_2 e_2$ be their separation, developed inside the cut disc. Then:*

(i) $s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \varepsilon_1 d_1 + \varepsilon_2 d_2$ with signs $\varepsilon_1, \varepsilon_2 \in \{+1, -1\}$ that depend on the configuration but not on ℓ, and $|B_A| = \sqrt{2}\,|d|$. The two turns can close together only if c_5 lies, as seen from c_3, on one particular diagonal of the field.

(ii) If c_3 and c_5 lie on one straight line of the field inside A, of length δ (in a T-mesh: they are joined by a chain of arcs along one trace, and δ is the sum of their lengths), then $|s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell)| = \delta$. The two turns can both be rings only if $\delta = 0$, that is if the two singularities merge; as long as the bounds of separation keep $\delta \geq 1$, one of them at least is a helix.

(iii) For whole lengths, $s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) \equiv d_1 + d_2 \pmod 2$: a pair at an odd number of edges from one another, counted along the two directions of the field, never lies between two rings.

Proof. With $R_3 = R$ and $R_5 = R^{-1}$, the sum of Proposition 3 for the two possible orders is $B_A = R^{-1}(I - R)\, p_3 + (I - R^{-1})\, p_5 = (I - R^{-1})\, d$ or $B_A = R\,(I - R^{-1})\, p_5 + (I - R)\, p_3 = (I - R)(-d)$. In coordinates (e_1, e_2), $R(a, b) = (-b, a)$, so $(I - R^{-1})(d_1, d_2) = (d_1 - d_2,\ d_1 + d_2)$

and $(I-R)(-d_1,-d_2) = (-d_1-d_2,\ d_1-d_2)$; reversing the orientation of the border, which inverts the rotations, gives the two other sign patterns. (We checked the four cases symbolically.) In all of them both components of B_A are of the form $\pm d_1 \pm d_2$ and $|B_A|^2 = 2|d|^2$, which is (i); the component along e_2 vanishes exactly on the line $\varepsilon_1 d_1 + \varepsilon_2 d_2 = 0$, a diagonal. For (ii), choose κ away from the straight segment joining c_3 to c_5, which does not separate the two borders of A, take the tree of part (a) to contain that segment so that the positions of (i) are those of the first visit, and develop c_5 from c_3 along the segment: d is δ times one of $\pm e_1$, $\pm e_2$, so $|\varepsilon_1 d_1 + \varepsilon_2 d_2| = \delta$. For (iii), $d \in \mathbb{Z}^2$ and $\pm d_1 \pm d_2 \equiv d_1 + d_2$. □

Plain counting confirms (iii) independently. If both turns were rings, the along component of B_A would be the difference $n_1 - n_2$ of the numbers of edges round them, of the parity of $d_1 + d_2$; but an annulus made of quads has an even number of edges on its border, since $4F = 2E_{\text{inside}} + E_{\text{border}}$, so $n_1 + n_2$ is even. Proposition 6 below makes this a statement about each turn. Part (ii) is the mechanism the problem of feasibility in Section 11 guessed, and Proposition 7 extends it from straight chains to staircases. It does not explain the conflict found on the male body at a single arc of the left wrist (Section 7): that conflict involves a single turn, not two (Remark 4), and Proposition 9 gives its mechanism.

For more singular vertices between two turns, the parity part generalizes into a checkerboard rule.

Proposition 5 (Checkerboard rule)**.** *In the setting of Proposition 3(b), with whole lengths, colour the points $p \in \mathbb{Z}^2$ of the cut disc black or white by the parity of $p_1 + p_2$ (either colour may be called white). Then $s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell)$ has the parity of the number of singular vertices of odd valence inside A that develop onto white points. In particular, two rings bound an annulus only if that number is even.*

Proof. In the sum of Proposition 3, a vertex of even valence has $R_i = I$ or $R_i = -I$, and $(I - R_i)p_i \in \{0, 2p_i\}$ is even. A vertex of odd valence has $R_i = R^{\pm 1}$, and both components of $(I - R^{\pm 1})(a,b) = (a \pm b,\ b \mp a)$ have the parity of $a + b$. Quarter-turns keep the property "both components have the parity σ", and sums add the σ. So both components of B_A have the parity of the number of odd vertices on white points. The count does not depend on the coordinates: a quarter-turn keeps colours, and a whole translation t changes the colour of every point if $t_1 + t_2$ is odd, which keeps the parity of the count because the number of odd vertices inside A is even, their indices $4 - k$ being odd and summing to zero with the even ones (Lemma 1); for the same reason the two colours give counts of the same parity. Developing a vertex along another path inside the cut disc moves it by a rotation $x \mapsto R_j(x - p_j) + p_j$ about another singular vertex, which keeps its colour. □

We also checked the rule on 13,000 random configurations with SymPy.

Parity on a sphere. The checkerboard rule compares two turns. On a closed surface of genus zero each turn carries its own parity condition, which needs neither a simple curve nor the holonomy. Let B be the matrix of the balances of the patches, so that the balanced lengths are the solutions of $B\ell = 0$.

Proposition 6 (Parity of a turn on a sphere)**.** *Let the surface be closed of genus zero and the T-mesh admit lengths that are positive and balance every patch (its real lengths do). For every turn γ and all whole lengths ℓ balancing every patch, of any sign,*

$$\alpha_\gamma(\ell) \equiv s_\gamma(\ell) \pmod 2.$$

Proof. First let ℓ be whole, balanced, and positive on every arc. The patches are then grids of unit squares glued along their arcs, a quad mesh of the sphere. Its graph is bipartite: over $\mathbb{Z}/2$, the cycles of a graph embedded in a sphere are sums of face boundaries, since $H_1(S^2; \mathbb{Z}/2) = 0$,

and every face boundary has four edges, so every cycle has an even length. Fix a colouring of the vertices in two colours, adjacent vertices differing. In each patch P the grid is connected, so the colour of its point of local coordinates (i, j) is $i + j + \kappa_P \bmod 2$ for a constant κ_P. For a motion $M(x) = R^k x + t$ with $t \in \mathbb{Z}^2$, write $\pi(M) = t_1 + t_2 \bmod 2$; since a quarter-turn keeps the parity of $i + j$, the sum of the coordinates of $M(x)$ has the parity of that of x plus $\pi(M)$.

Let $P_0, P_1, \ldots, P_N = P_0$ be the patches crossed by γ and D_k the motion that develops the coordinates of P_k into those of P_0, as in the proof of Proposition 1; D_0 is the identity and D_N the translation $T_\gamma(\ell)$. We show $\kappa_{P_k} \equiv \kappa_{P_0} + \pi(D_k)$ by induction. A vertex w of the arc crossed from P_k to P_{k+1} has coordinates y in P_k and y' in P_{k+1} with $D_k(y) = D_{k+1}(y')$; its colour is $y_1 + y_2 + \kappa_{P_k} \equiv (D_k y)_1 + (D_k y)_2 + \kappa_{P_0}$ by the induction hypothesis, and also $y'_1 + y'_2 + \kappa_{P_{k+1}}$; as $(D_{k+1}y')_1 + (D_{k+1}y')_2 \equiv y'_1 + y'_2 + \pi(D_{k+1})$, this gives $\kappa_{P_{k+1}} \equiv \kappa_{P_0} + \pi(D_{k+1})$. For $k = N$ the patch is P_0 again, so $\pi(T_\gamma(\ell)) \equiv 0$. In the frame (e_1, e_2) of the first side, which is that of P_0 up to a quarter-turn and signs (the coordinates swap when the first side is along the first axis of P_0), $T_\gamma(\ell) = (\alpha_\gamma(\ell), \pm s_\gamma(\ell))$; a quarter-turn keeps the parity of the sum of the coordinates, hence $\alpha_\gamma(\ell) + s_\gamma(\ell)$ is even.

For general whole balanced ℓ: the balanced lengths form a rational subspace, positive real balanced lengths exist, and positivity is an open condition, so there are positive rational balanced lengths, and, multiplying by a common denominator, positive whole ones p. For N large, $\ell + Np$ is whole, balanced and positive, and $\alpha_\gamma + s_\gamma$, being linear, takes at $\ell = (\ell + Np) - Np$ the difference of two even values. □

The proof uses only that the quad graph is bipartite, so it fails in higher genus: on a torus, a 3×3 grid of quads has rings of three quads; and with borders (Theorem 1). On a closed sphere it gives at once:

Corollary 1. *On a closed surface of genus zero:*

(a) every ring of quads has an even number of quads;

(b) a helix of odd pitch winds round an odd number of quads per turn;

(c) closing a helix of odd pitch changes the length of its walk by an odd number of edges: a repair that keeps the length of a turn cannot close it if its pitch is odd, and changing the pitch by an odd amount always changes the length.

Proof. (a) By Proposition 2 a ring is the strip of a walk with $s = 0$, so α is even. (b) and (c) are the proposition read before and after the change. □

Part (a) also follows directly from the bipartite graph: either border of a ring is a cycle of the quad graph with as many edges as the ring has quads. It is the classical bipartiteness of quadrangulations of the sphere; more generally, homologous cycles of a quadrangulation have the same parity [31, Lemma 3.1]. The proposition is consistent with the earlier statements and refines them. For a turn whose curve bounds a disc D, the argument of the checkerboard rule applied to Proposition 3(a) shows that both components of $B_D = (\alpha_\gamma, s_\gamma)$ have the parity of the number of vertices of odd valence of D on white points: they are equal, which is Proposition 6 for that turn. For two parallel turns, the proposition applied to each gives $s_{\gamma_1} - s_{\gamma_2} \equiv \alpha_{\gamma_1} - \alpha_{\gamma_2}$, the two components of B_A, which the checkerboard rule and Proposition 4(iii) evaluate from the singularities: two rings ($s_{\gamma_1} = s_{\gamma_2} = 0$) have even lengths, so no 3–5 pair at odd separation, and no odd number of odd vertices on white points, lies between them. On the eight systems exported for the male body (Section 7), the proposition holds on all 1,838 ways round found by walking every strip of the meshes (measured).

Remark 2 (Borders; a second proof)**.** *The proposition and part (a) of the corollary need the surface to be closed: on a sphere with two border loops, an open cylinder of three quads round has a ring of three quads. The proof uses $H_1(S^2; \mathbb{Z}/2) = 0$, which borders break. On a closed*

sphere there is a second proof, which needs neither the bipartite graph nor positive lengths, and which the formalization of Section 10 found. Every loop of Σ° is a product of the loops λ_i round the singular vertices (Section 8.3), whose holonomies are quarter-turns about the points p_i. For whole lengths these are points of $\mathbb{Z}^2$, and a quarter-turn about a whole point keeps the parity of $p_1 + p_2$. So the holonomy of every loop keeps that parity, and its translation, the image of the origin, has an even sum of coordinates: for a turn, $\alpha_\gamma + s_\gamma$ is even. Theorem 1 gives the rule on any surface, and the eye sockets of Suzanne a case met in practice: a ring of 23 *quads.*

How many independent rings. Rings that cross neither one another nor themselves are disjoint simple closed curves of Σ°, none bounding a disc with fewer than two singular vertices (Lemma 1). On a closed surface of genus zero with $n \geq 3$ singular vertices there are at most $n-3$ such curves pairwise non-homotopic, the number of curves of a decomposition of the n-punctured sphere into pairs of pants, the classical bound on the number of cylinders of a Jenkins–Strebel differential [53]; disjoint homotopic ones bound an annulus without singular vertex and share their shift by Proposition 3(c). The rings of a quad mesh that do not cross therefore fall into at most $n-3$ cylinders, the cylinders of the Jenkins–Strebel picture, and their closing conditions have rank at most $n-3$. On a surface of genus g with b border components the count of a decomposition into pairs of pants is $3g-3+n+b$, plus at most b classes of collars parallel to the borders (Proposition 14); a torus without singular vertex has one class.

Remark 3 (Whole periods are automatic; rings need zero)**.** *For whole lengths, every shift is a whole number: the period of the integer grid along a loop is always whole, which is what quantization guarantees and what [33] enforces by rounding. Closing a ring is the stronger requirement that this period be zero. The real shift of the parametrization is the twist of the field's cylinder; it tells which whole value rounding is likely to pick, not whether zero is reachable.*

Remark 4 (Sign-definite turns; Farkas certificates)**.** *Say that a linear form f in the lengths equals g* modulo the balances *if $f-g$ is a combination of the rows of B, so that $f(\ell) = g(\ell)$ for every balanced ℓ. If the shift of a turn equals, modulo the balances, a form $\sum_a c_a \ell_a$ with every $c_a \geq 0$, then for all balanced $\ell \geq 0$ the shift $s_\gamma(\ell)$ is never negative, and it is zero only if every arc with $c_a > 0$ has length zero. If one of those arcs is held open by a bound of separation, the turn is a helix under every quantization meeting the bounds; it should never be held, and one linear program per turn finds such turns. More generally, by Farkas' lemma (with $x = b + z$, $z \geq 0$), the system $B\ell = 0$, $s_\gamma(\ell) = 0$ for $\gamma \in H$, $\ell \geq b$ has no solution exactly when some combination $\sum_{\gamma \in H} u_\gamma s_\gamma$ equals, modulo the balances, a form $\sum_a c_a \ell_a$ with $c \geq 0$ and $\sum_a c_a b_a > 0$; and an arc a has length zero in every solution with $\ell \geq 0$ exactly when some such combination has $c_a > 0$. The data being integral, bounds included, the multipliers of the shifts and of the balances can be taken whole.*

On the male body this is what happens at the left wrist (measured; the certificate checked in exact rational arithmetic for way 0): in ways 0, 1, 2 and 4 one held turn, of length 7.0, has a shift equal, modulo the balances, to $-(\ell_{1921} + \ell_{1937})$, where arc 1937, 0.08 edges long, is held at length at least one. That turn cannot be closed on its own, within the engine's bounds of separation. Whether $\ell_{1937} = 0$ would merge two singular vertices, that is whether every valid quantization meets this bound, we did not check. Section 8.2 shows where this form comes from.

Remark 5 (Whole solutions exist at some size)**.** *The balance and closing constraints are homogeneous and the bounds of separation are lower bounds $\ell \geq b$ with $b \geq 0$. If they have a real solution they have a rational one, a vertex of a rational polyhedron, and multiplying it by the common denominator N gives whole lengths $N\ell \geq Nb \geq b$ that satisfy them all. Closing in whole numbers is thus never impossible in itself; what can fail is closing* at a given size *(number of quads) or within a given deviation from the targets. Appendix F measures this on the male body.*

On the male body. Appendix B tests the statements above on the T-mesh of the male body exported in Section 7: they hold on every turn and annulus they apply to. It also shows that most of the turns held for closing have curves that cross themselves, so that the statements on simple curves say nothing about them; Section 8.3 handles any curve.

8.2 Conflicts between turns

Conflicts are local, but not always pairs. One may hope that whenever a set of turns, each closable on its own, cannot be closed together within the bounds of separation, it contains two parallel turns whose annulus forbids it, as in Proposition 4. This is false. We searched the held turns of the eight exported systems for *irreducible* infeasible sets: sets that cannot be closed together within the bounds while every proper subset can (Table 4). For each, the Farkas multipliers of Remark 4 were made rational and the identity checked in exact arithmetic, and for each turn left out, whole lengths closing the others within the bounds were found by HiGHS and checked in exact integer arithmetic (measured).

ways	size	turns	multipliers	bounded arcs	exact check
0, 1, 2, 4	1	42 (way 0)	-1	1937	way 0
1, 3, 5, 7	2	54, 61 (way 1)	$1, -1$	1587	—
1, 3	2	45, 49 (way 1)	$1, -1$	1069	—
1, 3, 7	2	26, 37 (way 1)	$-1, 1$	619	—
5, 7	2	58, 94 (way 5)	$1, -1$	858	—
5	2	49, 71	$-1, 1$	619, 625	—
6	2	57, 58	$1, 1$	1241	—
7	2	8, 15	$1, 1$	191	—
1	3	36, 37, 39	$1, 1, -1$	619	yes
1	4	40, 56, 59, 61	$1, 1, -1, -1$	1587	yes
1	4	66, 71, 75, 76	$-\frac{7}{6}, -\frac{5}{6}, -\frac{1}{3}, -\frac{1}{2}$	1967, 1980, 1998	yes
3	4	73, 78, 82, 83	$-\frac{7}{4}, -\frac{5}{4}, -\frac{1}{2}, -\frac{3}{4}$	1980, 1998	yes
5	7	59, 91, 96, 98, 99, 157, 158	$2, 4, 1, -4, -1, 1, -1$	1036	yes
7	7	112–116, 145, 147	$8, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}$	1157	yes
5	15	0, 2–9, 11, 14, 16–18, 64	whole, up to 23	17	14 of 15

Table 4: Irreducible infeasible sets of held turns on the male body, bounds of separation kept. Turn numbers are those of the first way listed; "bounded arcs": arcs held at length at least one in the support of the non-negative form. Rows without an exact check come from the floating-point linear program; their turns are each closable alone, so a pair is irreducible. In the last row, the whole lengths closing the set without turn 3 were not found within the time limit.

The smallest counterexample has three turns, in way 1 near the right eye, of lengths 54.9, 52.8 and 108.1:

$$s_{36} + s_{37} - s_{39} = \ell_{604} + \ell_{619} + \ell_{620} + \ell_{637} \quad \text{modulo the balances,}$$

with $\ell_{619} \geq 1$ imposed, so no lengths meeting the bounds close all three, while every two of them close together in whole numbers. Sets of four and seven turns behave alike, with the same signature: a combination of the shifts equals a non-negative form containing a bounded arc. Every set found is infeasible in real numbers already: none of these obstructions is one of parity. In the conflicts between two turns, the sum or difference of the shifts equals a sum of two to eight arc lengths with coefficient one, one of them bounded; the two turns have nearly equal lengths (39.7 and 40.0, 52.4 and 52.8, 277.5 and 278.0, ...) and lie in clusters of singularities, at the eyes, the hands and the feet. One may read this as Proposition 4(ii): a vertex of valence three and one of valence five joined along one trace by a chain holding the bounded arc. The T-mesh refutes that reading: we examined 13 conflicts between two turns (every row of two

turns in Table 4 except the pair 49, 71 of way 5), and in none of them are the turns separated by such a pair on one trace. What separates them is a staircase.

Proposition 7 (Staircases)**.** *Let γ_1, γ_2 be turns and c, c' singular vertices such that*

$$s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \varepsilon_1 d_1 + \varepsilon_2 d_2$$

for all balanced ℓ, with $\varepsilon_1, \varepsilon_2 \in \{+1, -1\}$ and $d = p_{c'} - p_c = d_1 e_1 + d_2 e_2$ the separation of c' from c developed along a path π from c to c' (Propositions 4(i) and 8 give such identities). Let σ be a path from c to c' along arcs $a_1, \dots, a_r$, through no other singular vertex, and homotopic to π with its ends fixed in Σ without the other singular vertices. Developed along σ, each arc a_i runs along a direction $v_i \in \{\pm e_1, \pm e_2\}$. If $\varepsilon_1 v_{i,1} + \varepsilon_2 v_{i,2} = 1$ for every i, so that σ is a staircase*, using two directions a quarter-turn apart and climbing across the diagonal on which the two turns could close, then*

$$s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \sum_{i=1}^{r} \ell_{a_i}$$

for all balanced ℓ. Both turns are rings only if every arc of σ has length zero. If this sum equals, modulo the balances, a form with non-negative coefficients containing an arc held at length at least one, one of the two turns is a helix under every quantization meeting the bounds.

Proof. Developing along σ adds, arc after arc, the vector $\ell_{a_i} v_i$: the nodes σ passes are T-junctions, where the flat metric has no cone point. As σ and π are homotopic with fixed ends away from the other cone points, they develop c' at the same place relative to c, so $d = \sum_i \ell_{a_i} v_i$ and $\varepsilon_1 d_1 + \varepsilon_2 d_2 = \sum_i \ell_{a_i}$, first for positive balanced lengths, then for all by linearity. If both shifts vanish, this sum of non-negative lengths is zero. □

The proof uses only $\varepsilon_1 v_{i,1} + \varepsilon_2 v_{i,2} = 1$ for each step: neither that the steps run along the axes nor that $\varepsilon_1, \varepsilon_2 = \pm 1$, so it applies to any path on whose steps the linear form of the shift difference is one.

Proposition 4(ii) is the case of a straight σ, along one direction. The second source of such identities is an exchange of two vertices of equal valence.

Proposition 8 (Exchanging two vertices)**.** *Let γ_1, γ_2 be turns, and κ a path from the start of $\bar{\gamma}_1$ to that of $\bar{\gamma}_2$ that carries the walking direction of γ_2 onto that of γ_1, as in Proposition 3(b); let $\lambda = \bar{\gamma}_1 \, \kappa \, \bar{\gamma}_2^{-1} \kappa^{-1}$. Suppose that λ is homotopic in Σ° to $\lambda_c \lambda_{c'}^{-1}$, where λ_c and $\lambda_{c'}$ are small loops run with c, c' on their right round two singular vertices of the same valence k, reached along paths that develop them at p_c and $p_{c'}$. Then*

$$s_{\gamma_1}(\ell) - s_{\gamma_2}(\ell) = \langle e_2, \ (I - R^{4-k})(p_c - p_{c'}) \rangle.$$

For valence three or five this is $\pm d_1 \pm d_2$ with $d = p_c - p_{c'}$, as for a dislocation, and Proposition 7 applies.

Proof. As in the proof of Proposition 3(b), $\hat{h}(\lambda)$ is the translation by $\hat{h}(\bar{\gamma}_1) - \hat{h}(\bar{\gamma}_2)$, read in the coordinates at the start of γ_1. With $j = 4 - k$,

$$\hat{h}(\lambda_c) \circ \hat{h}(\lambda_{c'})^{-1}(x) = R^j \big(R^{-j}(x - p_{c'}) + p_{c'} - p_c\big) + p_c = x + (I - R^j)(p_c - p_{c'}),$$

and the shifts are the components along e_2. For $j = \pm 1$ both components of $(I - R^j)d$ are of the form $\pm d_1 \pm d_2$, as in the proof of Proposition 4. □

Table 5 lists the six configurations the 13 conflicts between two turns fall into. Configuration A is a dislocation: the two turns are simple and disjoint, and the annulus between them holds one vertex of valence three and one of valence five, 0.270 and 0.720 edges apart across and

along, developed in the cut annulus. The difference of the shifts is $d_1 - d_2$ up to sign, as Proposition 4(i) says, and the staircase of two arcs joining the pair has length 0.990, the difference of the real shifts -0.955 and -1.945 of the two turns, identically on balanced lengths; modulo the balances it is a non-negative form containing the held arc 619. At the whole lengths the engine found, the two vertices are one edge apart on one grid line, the case of Proposition 4(ii) with $\delta = 1$. In configurations B, C and D the two turns cross. The combination of their curves winds $+1$ round a region holding one vertex of valence three, -1 round another holding another such vertex, and round no other singular vertex: in homology, an exchange of two vertices of valence three. The combination of the shifts is the length of a staircase joining them, and the held arc lies on it (D) or follows from it by the balances (B, C). The homotopy that Proposition 8 assumes was not checked; the identity it leads to with Proposition 7 was. In configurations E and F, the combination winds once round singular vertices of total index four, like a turn round a cap; a bounded search found no staircase, nor sum of two, equal to the combination. Section 8.4 shows what they are: closed chains of six and of four combs, neither of which is a non-negative combination of staircases, caps and winding separatrices.

	conflict	curves	between the turns	staircase	mechanism
A	$s_{37} - s_{26}$, ways 1, 3, 7; right eye	disjoint, simple; 52.4 and 52.8	annulus: 78 (val. 3), 76 (val. 5)	2 arcs, 0.990; held arc 619 by the balances	dislocation
B	$s_{45} - s_{49}$, ways 1, 3; right hand	simple; 39.7 and 40.0; 2 crossings	lenses: 102, 103 (val. 3)	4 arcs, 1.731; held arc 1069 by the balances	exchange
C	$s_{58} - s_{94}$, ways 5, 7; right hand	2 self-crossings each; 278.0 and 277.5; 6 crossings	109, 118 (val. 3)	3 arcs, 2.203; held arc 858 by the balances	exchange
D	$s_{57} + s_{58}$, way 6; left foot	figure-eights; 236.4 and 236.9; 4 crossings	155, 156 (val. 3)	2 arcs, 1.452, one of them held	exchange
E	$s_{54} - s_{61}$, ways 1, 3, 5, 7; left eye	simple and figure-eight; 59.5 and 71.9; 2 crossings	11 vertices of index 3, and 207 (val. 3)	none; a closed chain of 6 combs	chain of combs
F	$s_8 + s_{15}$, way 7; right foot	2 and 1 self-crossings; 251.5 and 247.4; 4 crossings	14, 23, 24, and 15 (val. 3)	none; a closed chain of 4 combs	chain of combs round a cap

Table 5: The conflicts between two turns on the male body, by configuration (measured). Turn numbers are those of the first way listed, signs those of the exported shifts; "between the turns": the singular vertices round which the combination of the two curves winds, numbered as nodes of the T-mesh; "staircase": a staircase between them whose length equals the combination of the shifts modulo the balances, and where the held arc of the certificate lies. A: Proposition 4(i) with Proposition 7; B–D: Proposition 8 with Proposition 7; E, F: Section 8.4.

A separatrix that winds. The one turn that cannot be closed on its own is of another kind.

Proposition 9 (A separatrix that winds)**.** *Let σ and τ be separatrices of a singular vertex v such that σ, followed from v, stops on τ at a point q at distance $\delta(\ell)$ from v along τ, a sum of arc lengths. Let λ be the loop that runs along σ from v to q and back along τ to v, pushed off v into the sector it turns through. If λ is freely homotopic in Σ° to the curve $\bar{\gamma}$ of a turn γ, the walking direction of γ going to that of σ, then $s_\gamma(\ell) = \pm\delta(\ell)$ for all balanced ℓ. The*

turn is a ring only if $\delta = 0$, that is if σ comes back onto its own vertex; if δ equals, modulo the balances, a non-negative form containing an arc held at length at least one, γ is a helix under every quantization meeting the bounds.

Proof. Develop λ from a point of σ next to v, with v at the origin and σ along e_1. The separatrix σ is straight and stops on τ at a right angle, so the return along τ is a segment of length δ along $\pm e_2$, like the closing segment of a turn. Being homotopic to the curve of a turn, λ has a translation as holonomy; it maps v, developed at the start, onto v developed at the end, which lies at distance δ from q across σ. Its component across is thus $\pm\delta$. A free homotopy conjugates the holonomy by a motion; the conjugate of a translation is the translation by the rotated vector, and the rotation is the identity since it carries the walking direction of γ onto that of σ. □

At the left wrist the held turn (42 in way 0, length 6.99, real shift -0.305, whole shift -1) is simple, and its small side holds four vertices of valence three, 226, 243, 244 and 245, of total index 4: a cap. Proposition 3(a) on that side gives the shift -0.305 from their four positions, but as a form in the lengths of the arcs inside the cap it has no sign. The sign comes from outside. The vertex 246, of valence five, lies beyond the cap; one of its separatrices, the trace 964, runs $0.066 + 6.928 = 6.994$ edges once round the wrist, parallel to the turn, and stops on another of its separatrices, the trace 966, at the T-junction 1099, at the distance $\delta = \ell_{1920} + \ell_{1921} = 0.075 + 0.230 = 0.305$ from 246. The shift of the turn is $-\delta$ modulo the balances, and ℓ_{1920} equals, modulo the balances, the length of arc 1937, which starts at the vertex 248 of valence five and is held at length at least one: whence the certificate $-(\ell_{1921} + \ell_{1937})$ of Remark 4, and $s \leq -1$ at every admissible whole length (measured). A ring would need the separatrix to come back onto its own vertex, which the engine's bounds of separation forbid. As a check, the along component of the holonomy is the length of σ, 6.994, and that of the turn is 6.99. We checked the identity, not the homotopy that Proposition 9 assumes.

Conflicts of more than two turns. The three turns at the right eye are a dislocation between three turns. The turns 36 and 37 are simple; 39 is a figure-eight; each two of them cross twice, and together they cut the surface into five regions. The cycle $\bar{\gamma}_{36} + \bar{\gamma}_{37} - \bar{\gamma}_{39}$ winds -1 round the one region bordered by all three curves, which holds the vertices 79 of valence three and 77 of valence five, and round no other singular vertex: in homology, 39 is the sum of 36 and 37 and a loop round the pair. The combination $s_{36} + s_{37} - s_{39}$ is the length of a staircase of three arcs from 79 to 77 that includes the held arc 619, $1.367 = 1.712 - 0.955 + 0.610$ at the real lengths (measured). The set of four turns at the eyes (40, 56, 59, 61 of way 1, 72 to 167 long, crossing themselves up to 17 times and each other up to 28 times) winds $+1$ round the region holding 198 (valence three) and 199 (valence five), -1 round 207 and $+1$ round 202 (both valence three), and round no other singular vertex: in homology, a dislocation and an exchange, the held arc 1587 starting at 199; a bounded search did not write the combination as a sum of two staircases, and a floating-point test of Section 8.4 separates its certificate from the non-negative combinations of staircases, caps and winding separatrices.

The fractional multipliers are an artefact of normalization. The linear program that finds the certificate asks the bounded arcs to weigh one in total: at the left hand the form has three bounded arcs of coefficient $\frac{1}{3}$ in way 1 and two of coefficient $\frac{1}{2}$ in way 3. Scaled to whole numbers, both ways give the same primitive multipliers,

$$7\,s_{66} + 5\,s_{71} + 2\,s_{75} + 3\,s_{76} = -2F \quad \text{modulo the balances,}$$

with F a form of non-negative whole coefficients 1, 2 and 5 containing the bounded arcs (measured). With these weights the weighted cycle winds alike round the rest of the body (236 singular vertices) and round the region of ten singular vertices of total index -2. Section 8.3 explains part of these weights: why $-2F$ is even, and which dislocation carries the coefficient 5.

Question 1 (Are conflicts round a cluster?)**.** *On a closed surface of genus zero, let H be an irreducible set of turns that cannot be closed together within the bounds of separation. Are the curves of the turns of H, up to the cylinders of Proposition 3(c), the $k \geq 1$ borders of one planar region round a cluster of singularities, and is a combination of their shifts that certifies the conflict given by the holonomy relation of these borders: a form with non-negative coefficients containing an arc held open?*

Answer: no. The certificates alone suggest it, and the T-mesh confirms it for one turn (the wrist, a cap, though the form comes from a separatrix outside the region) and for the annulus of configuration A. It holds in a modified form for the three turns at the right eye: the curves are not the borders of a pair of pants, 39 being a figure-eight, but their combination winds round one region holding one pair of valences three and five, and the certificate is the staircase of that pair. As stated, it is refuted by the conflicts made of crossing turns: configurations B to F, and the sets of four and seven turns, whose curves cross each other up to tens of times and bound no single planar region. What the confirmed cases share is local, which suggests a refined question.

Question 2 (Are conflicts sums of local dislocations?)**.** *On a closed surface of genus zero, let H be an irreducible set of turns that cannot be closed together within the bounds of separation. Does some combination $\sum_{\gamma \in H} u_\gamma s_\gamma$ with whole multipliers that certifies the conflict (Remark 4) equal, modulo the balances, a sum with non-negative whole weights of*

(i) lengths of staircases (Proposition 7) between a vertex of valence three and one of valence five, or between two vertices of equal valence exchanged as in Proposition 8, and

(ii) shifts of caps: sign-definite forms carried by curves that wind once round singular vertices of total index four, as the wrist turn and its separatrix (Proposition 9),

one term at least containing an arc held open?

Answer: no as stated, yes and empty when read loosely. It holds, with a single term and up to the homotopies we did not check, for the wrist, configurations A to D and the three turns at the right eye. Section 8.4 settles it on the male body: read strictly, with staircases of weight one, monotone caps and winding separatrices only, it is false (configurations E and F, in exact arithmetic); read loosely, with every local dislocation allowed, it is true for every certificate and says nothing that Farkas' lemma does not.

8.3 Any combination of turns

The conflicts above were described by the winding numbers of a weighted cycle $\sum_\gamma u_\gamma \bar{\gamma}$. Winding numbers are its class in the homology of Σ°, which is commutative, while the holonomy is not; so they cannot carry the whole holonomy, and Proposition 3 needed an order of first visit. The bookkeeping that keeps the order is the free differential calculus of Fox [23]. It turns Proposition 3 into a formula for any combination of turns.

Setting. Let Σ be closed of genus zero, with singular vertices $c_1, \ldots, c_n$ (Proposition 15 adds the terms of borders and handles). Fix a point x_0 inside a patch, with the frame of that patch, and disjoint paths π_i from x_0 to c_i; let λ_i be the loop that follows π_i, goes round c_i with c_i on its right, and comes back. The fundamental group of Σ° is generated by the λ_i, with a single relation, the product of all of them in the order in which the paths leave x_0. Let p_i be the developed position of c_i at the end of π_i and $R_i = R^{4-k(c_i)}$, so that $\hat{h}(\lambda_i) = \rho_i$, $\rho_i(x) = R_i(x - p_i) + p_i$. For a loop γ at x_0 write $\hat{h}(\gamma)(x) = \rho(\gamma)x + \tau(\gamma)$; since $\hat{h}$ is a homomorphism, $\tau(\alpha\beta) = \tau(\alpha) + \rho(\alpha)\,\tau(\beta)$. Let F be the free group on letters $x_1, \ldots, x_n$, mapped onto the fundamental group by $x_i \mapsto \lambda_i$, and $\partial/\partial x_i$ the Fox derivatives of its group ring: $\partial x_j/\partial x_i = \delta_{ij}$ and $\partial(uv)/\partial x_i = \partial u/\partial x_i + u\,\partial v/\partial x_i$. Extend ρ linearly to the group ring; its values are the

integer matrices $a_0 I + a_1 R + a_2 R^2 + a_3 R^3$, and the augmentation ϵ maps such an element to $a_0 + a_1 + a_2 + a_3$, the sum of the coefficients of the group ring element.

The first identity below is the defining property of the Fox derivatives for a crossed homomorphism of the free group [23], applied to the translation part of the holonomy, with $\tau(\lambda_i) = (I - R_i)p_i$; the developed positions of the cone points are, up to a common motion, the twisted-cohomology coordinates of the flat metric [55, 57]. What is new is its reading on combinations of turns.

Proposition 10 (Fox–Burgers formula)**.** *For every word w of F,*

$$\tau(w) = \sum_{i=1}^{n} \rho\Big(\frac{\partial w}{\partial x_i}\Big)(I - R_i)\, p_i.$$

Let $\gamma_1, \dots, \gamma_m$ be turns, κ_r a path from x_0 to the start of $\bar\gamma_r$, f_r the direction e_2 of γ_r carried back to x_0 along κ_r (one of $\pm e_1$, $\pm e_2$), and w_r a word for $\kappa_r \bar\gamma_r \kappa_r^{-1}$. For multipliers u_r define the twisted winding *of the weighted cycle round c_i,*

$$W_i = \sum_{r=1}^{m} u_r\, \rho\Big(\frac{\partial w_r}{\partial x_i}\Big)^{\top} f_r \in \mathbb{Z}^2 \ \textit{(for whole } u_r\textit{)}.$$

Then, for all balanced lengths,

$$\sum_{r=1}^{m} u_r\, s_{\gamma_r}(\ell) = \sum_{i=1}^{n} \big\langle W_i,\ (I - R_i)\, p_i \big\rangle, \qquad \sum_{i=1}^{n} (I - R_i)^{\top} W_i = 0,$$

so the sum does not change when all the p_i are moved by one translation: it is a combination of separations $p_i - p_j$. The ordinary winding number of $\sum_r u_r \bar\gamma_r$ round c_i is, up to a constant common to all i, $w_i = \sum_r u_r\, \epsilon(\partial w_r / \partial x_i)$. The words w_r are defined up to the relation, and the W_i accordingly: inserting a conjugate $g\varrho g^{-1}$ of the relation ϱ in w_r, after a prefix a of w_r, adds $u_r \rho(ag\, \partial\varrho/\partial x_i)^{\top} f_r$ to every W_i (with a empty, $u_r \rho(g\, \partial\varrho/\partial x_i)^{\top} f_r$), and leaves the sum unchanged.

Proof. The map $w \mapsto \tau(w)$ and the right-hand side are both maps $F \to \mathbb{R}^2$ with $\phi(uv) = \phi(u) + \rho(u)\phi(v)$: the first by the rule above, the second by the rule of the Fox derivatives and the multiplicativity of ρ. They agree on the letters, where $\partial x_j / \partial x_i = \delta_{ij}$ and $\tau(\lambda_i) = (I - R_i)p_i$, and on the empty word, so they agree everywhere by induction on the length of words (for x_i^{-1}, use $0 = \phi(x_i x_i^{-1})$). For a turn, $\hat h(\kappa_r \bar\gamma_r \kappa_r^{-1})$ is the conjugate of the translation $\hat h(\bar\gamma_r)$ by $\hat h(\kappa_r)$, the translation by $M_r T_r$, where T_r is the translation of $\bar\gamma_r$ in its own frame and M_r the rotation of $\hat h(\kappa_r)$. Hence $s_{\gamma_r} = \langle e_2, T_r \rangle = \langle M_r e_2, M_r T_r \rangle = \langle f_r, \tau(w_r) \rangle$, and $\langle f, \rho(D)\, b \rangle = \langle \rho(D)^{\top} f, b \rangle$ gives the formula. The fundamental formula of Fox calculus, $\sum_i (\partial w / \partial x_i)(x_i - 1) = w - 1$, maps under ρ to $\sum_i \rho(\partial w_r / \partial x_i)(R_i - I) = \rho(w_r) - I = 0$, the rotation of a turn being trivial; transposed and summed with the $u_r f_r$, it is the second identity. Finally $\epsilon(\partial w / \partial x_i)$ is the exponent sum of x_i in w, which is the class of w in the homology of Σ° written on the basis of the λ_i, defined up to the relation, in which each λ_i appears once: up to a common constant, the winding numbers. □

For a simple turn the word is a product of conjugates of the x_i, each once, and the formula is Proposition 3(a), with $\rho(\partial w / \partial x_i) = Q_i$. In general the twisted winding W_i is an integer vector, not a number: the cycle may go round c_i several times, each time with another rotation before it. Reduced modulo 2, the rotations disappear.

Corollary 2 (Checkerboard rule for any combination)**.** *On a closed surface of genus zero, let ℓ be whole and balanced and the multipliers u_r whole. Colour each singular vertex c_i by the parity χ_i of $p_{i,1} + p_{i,2}$; it does not depend on the paths π_i, and for positive lengths it is the colour of c_i in the bipartition of the quad mesh. Then*

$$\sum_r u_r\, s_{\gamma_r}(\ell) \equiv \sum_r u_r\, \alpha_{\gamma_r}(\ell) \equiv \sum_{i:\ k(c_i)\ \textit{odd}} w_i\, \chi_i \pmod 2,$$

with w_i the winding numbers of $\sum_r u_r \bar{\gamma}_r$, whatever the common constant and whatever the orientation of each $\bar{\gamma}_r$. In particular, a combination whose cycle winds an even number of times round every vertex of odd valence takes even values at every whole length.

Proof. Write $\rho(D) = \sum_k a_k R^k$, with $\sum_k a_k = \epsilon(D)$. Since $(R - I)^2 = -2R$, $R^k(I - R^{\pm 1}) \equiv I - R^{\pm 1}$ modulo 2, so $\rho(D)(I - R_i) \equiv \epsilon(D)(I - R_i)$ for $k(c_i)$ odd, and $\rho(D)(I - R_i) \equiv 0$ for $k(c_i)$ even, where $I - R_i \in \{0, 2I\}$. For $f \in \{\pm e_1, \pm e_2\}$ and $p \in \mathbb{Z}^2$, $\langle f, (I - R^{\pm 1})p \rangle \equiv p_1 + p_2$. This gives the value modulo 2 of the sum of Proposition 10, and the same computation with the direction e_1 of each turn gives that of $\sum_r u_r \alpha_{\gamma_r}$. A rotation about a singular vertex at a whole point, or a quarter-turn, keeps the parity of $p_1 + p_2$, so χ_i does not depend on the paths; a whole translation of the origin flips all colours, which adds $\sum_{k(c_i) \text{ odd}} w_i$, even because the rotation of the cycle is trivial ($\sum_i w_i(4 - k(c_i)) \equiv 0 \pmod 4$). Adding a constant q to all w_i adds $q \sum_{k(c_i) \text{ odd}} \chi_i$, which is even: it is the value modulo 2 of the relation, a loop with zero translation. Reversing a turn changes the sign of its winding numbers, which does not matter modulo 2. For whole lengths with every patch side positive the developed parity is a proper colouring of the quad graph, as in the proof of Proposition 6. □

The corollary contains the checkerboard rule of Proposition 5 (winding one inside the annulus, zero outside), Proposition 4(iii), and, on a sphere, Proposition 6: for a single turn s and α have the same parity because both are congruent to the same sum. With borders or handles the colours may depend on the paths: the count of vertices on white points still gives the parity when every curve lies in a planar region without border, and in general $\sum_r u_r(\alpha_{\gamma_r} + s_{\gamma_r}) \equiv \sum_r u_r \, \Omega_\ell([\bar{\gamma}_r])$ (Theorem 1).

As in the theory of crystal defects, whose group of defects is not commutative [37], and as in the commutator argument of Izmestiev et al. [29, §7], winding numbers do not determine the holonomy:

Proposition 11 (Winding numbers do not suffice)**.** *Let c_a, c_b be singular vertices of odd valence developed at distinct points $p_a \neq p_b$. The loop $\lambda_a \lambda_b \lambda_a^{-1} \lambda_b^{-1}$ winds zero times round every singular vertex, and its holonomy is the translation by $(I - R_a)(I - R_b)(p_a - p_b)$, of length $2|p_a - p_b| \neq 0$. No function of the winding numbers gives the holonomy of the loops with trivial rotation.*

Proof. For $w = x_a x_b x_a^{-1} x_b^{-1}$, $\partial w / \partial x_a = 1 - x_a x_b x_a^{-1}$ and $\partial w / \partial x_b = x_a - w$, whose images under ρ are $I - R_b$ and $R_a - I$, the rotations commuting. By Proposition 10, $\tau(w) = (I - R_b)(I - R_a)p_a + (R_a - I)(I - R_b)p_b$. For odd valences $I - R_a$ and $I - R_b$ are $\sqrt{2}$ times a rotation. The exponent sums of w are zero. □

Measured on the male body. We implemented Proposition 10 on the T-mesh of Appendix B. The patches are developed along a breadth-first tree of patches; the arcs this tree does not cross form a spanning tree of the 1,295 nodes, and the surface cut along it is a disc with one chart. Crossing an arc of the tree changes the chart by a motion; rooted at a singular vertex, every such motion is the composition, in the cyclic order round the node, of the motions of the arcs below it and of the rotation about the node, the recursion in the proof of Proposition 3(a). It holds at every node of the tree with one convention of order, found among eight, the places p_i being corners of the nodes in the chart. The holonomy of every turn is then a word in the rotations, and the formula of Proposition 10 gives its translation, in both components, for all 1,299 turns of the eight ways (identities modulo the balances; measured throughout this paragraph). Corollary 2 holds, at the whole lengths of the engine, on all 1,299 turns, 718 simple, 101 with one self-crossing and 480 with more, and on the 14 infeasible sets of Table 4 with whole multipliers (all but the set of fifteen). For the set at the left hand, with multipliers $(7, 5, 2, 3)$, the cycle winds an even number of times round every vertex of odd valence: the corollary is why its certificate $-2F$ is even at every whole length.

Proposition 11 is not idle on this body: the turns 18 of way 0 (7 self-crossings), 22 of way 0 (simple), 101 of way 4 (4) and 65 of way 5 (17) satisfy $w_{18} = w_{22} + w_{101} + w_{65}$ exactly, round every singular vertex, while no choice of quarter-turn frames makes their translations satisfy the same relation (the smallest residual has norm 0.25 on an orthonormal basis of the balanced lengths). A "winding Burgers formula" with numbers for weights is thus false, on the body as in principle; it holds modulo 2 (Corollary 2), and over the reals with the twisted windings.

8.4 Combs

The twisted windings of Proposition 10 depend on the chart, so grouping them into clusters is not a statement about the surface. Another reading of a certificate is intrinsic, and turns every arc into a local dislocation.

Feet and combs. In the T-meshes of openclare, traced from the separatrices of the singular vertices only, every trace of a closed surface starts at a singular vertex (feature lines or seeded traces would break this; we assume it below). For an arc a, along the direction e_a from the node n_0 to the node n_1, call *foot* of n_j the node itself if it is singular, and otherwise the singular vertex where the trace through n_j perpendicular to a starts. The *comb* σ_a of a is the path from the foot c_0 of n_0 along its trace to n_0, along a to n_1, and along the other trace to the foot c_1 of n_1; it passes through T-junctions only. Call *graph of feet* the graph whose vertices are the singular vertices of odd valence and whose edges are the arcs whose two feet are distinct vertices of odd valence.

Lemma 3 (Combs)**.** *Let p_{c_0}, p_{c_1} be the feet of an arc a developed along its comb. Then, for all balanced lengths,*

$$\langle e_a,\ p_{c_1} - p_{c_0} \rangle = \ell_a.$$

If $c_0 \neq c_1$ have odd valence, $2\ell_a$ is, up to sign, the form of the twisted class $\lambda_{c_0}\lambda_{c_1}^{\pm 1} \otimes v$ of the loops round the two feet reached along the comb, for a whole v with $(I - R^{\pm 1})^\top v = 2e_a$: the pair is a dislocation if the valences differ (Proposition 4) and an exchange if they are equal (Proposition 8). If $c_0 = c_1$, ℓ_a is the shift of a winding separatrix (Proposition 9).

Proof. Developed along the comb, every step but a runs along a trace perpendicular to a and through no cone point, so it adds a vector orthogonal to e_a; the step along a adds $\ell_a e_a$. For the pair, the loop $\lambda_{c_0}\lambda_{c_1}^{\pm 1}$, with the sign that makes its rotation trivial for valences three and five and the inverse for equal valences, has the translation $\pm(I - R^{\pm 1})(p_{c_0} - p_{c_1})$, as in the proofs of Propositions 4 and 8; pairing it with v gives $\pm\langle (I - R^{\pm 1})^\top v, p_{c_0} - p_{c_1} \rangle = \pm 2\ell_a$. Such a v is whole because $I - R^{\pm 1}$ maps $\mathbb{Z}^2$ onto the lattice L of vectors whose two coordinates have the same parity, which contains $2e_a$. If $c_0 = c_1$, the two traces are separatrices of one vertex, the second stopping on the first at the distance ℓ_a, the setting of Proposition 9. □

A *closed walk of combs* is a sequence of arcs $a_1, \ldots, a_k$ whose feet are c_{i-1} and c_i, of odd valence, with $c_k = c_0$; its *reading* is $\sum_i \ell_{a_i}$. A walk of length one is a winding separatrix, a walk of length two between a vertex of valence three and one of valence five the staircase of Proposition 7, and a walk round four vertices of valence three a cap.

Proposition 12 (Certificates are sums of combs)**.** *Let $c = \sum_a c_a \ell_a$, $c_a \in \mathbb{Z}_{\geq 0}$, be a certificate (Remark 4) whose arcs have feet of odd valence and whose odd part, the arcs with c_a odd, is a cycle modulo 2 of the graph of feet. Then c is a sum with whole non-negative weights of comb readings $2\ell_a$, each the form of a dislocation or an exchange (Lemma 3), and of readings of closed walks of combs. With real weights, every non-negative form supported on arcs whose two feet are of odd valence, distinct or equal, is a non-negative combination of comb readings $2\ell_a$ and of winding separatrices ℓ_a, without condition of parity.*

Proof. The odd part, a cycle modulo 2, splits into closed walks along its edges (Veblen); what remains is $\sum_a k_a\, 2\ell_a$ with $k_a = \lfloor c_a/2 \rfloor$. With real weights, $c = \sum_a \frac{c_a}{2}\,(2\ell_a)$ on the arcs with two distinct feet, and $c_a\,\ell_a$, a walk of length one, on the arcs whose two feet are one vertex. □

Remark 6 (Closed walks as twisted cycles)**.** *We expect the reading of a closed walk of combs to be the Burgers form of a twisted cycle supported near its combs: by Lemma 3 and a summation by parts the reading is $\sum_i \langle e_{a_i} - e_{a_{i+1}},\ p_{c_i} \rangle$, and two unit vectors differ by an element of the image of $(I - R_{c_i})^\top$. But each p_{c_i} of Lemma 3 is developed along its own comb, and the two combs through c_i develop it in charts that differ by the rotation of the sector between them; a proof must develop the whole walk in one chart, insert these rotations, and account for the holonomy of the loop of the walk at its closing. We have not written it, and we use closed walks below only as forms in the lengths.*

Proposition 13 (The Burgers pairing)**.** *Let the surface be closed of genus zero, with no singular vertex of valence divisible by four, and let every trace of the T-mesh start at a singular vertex. Then the map that sends a class of the twisted homology $H_1(\Sigma^\circ; \mathbb{R}^2_\rho)$ to its form $\sum_i \langle W_i, (I - R_i)p_i \rangle$ on the balanced lengths is onto the dual of the balanced lengths. Its kernel contains the class across each separatrix connection. If some singular vertices have valence divisible by four, no form depends on their developed positions, and the map need not be onto.*

Proof. If every such form vanishes at balanced lengths ℓ, every loop of Σ° with trivial rotation has a holonomy without translation, since the components of its translation are such forms (Proposition 10). For two singular vertices a, b, the commutator of λ_a and λ_b has trivial rotation, the rotations commuting, and by the computation of Proposition 11 its translation is $(I - R_a)(I - R_b)(p_a - p_b)$. No R_i is the identity, so $I - R_i$ is invertible ($I - R^{\pm 1}$ is $\sqrt{2}$ times a rotation, $I - R^2 = 2I$), and $p_a = p_b$: every singular vertex develops at one point q. Every ρ_i fixes q; on a closed sphere the λ_i generate the fundamental group of Σ°, so every holonomy fixes q, and developed along any path every singular vertex lands at q. Every arc has feet, since every trace starts at a singular vertex, so every comb offset of Lemma 3 is zero, and $\ell = 0$. The image of the map is thus a subspace of the dual with no common zero but 0: the whole dual. Across a connection, a separatrix joins two singular vertices, whose separation along it is the length of the connection in the direction of the trace and zero across; the twisted class pairing the across direction with it has the zero form. A vertex of valence divisible by four has $R_i = I$, so $(I - R_i)p_i = 0$ and no form depends on p_i: its developed position is a relative period, which deformations of the lengths can move without changing the holonomy of any loop of Σ°. □

The pairing is the T-mesh form of a classical fact: the developed positions of the cone points are local coordinates of flat cone metrics, as twisted (co)homology classes [55, 57].

Measured on the male body. The T-mesh has 1,035 traces; 1,034 end at T-junctions and one is a separatrix connection, from vertex 204 to vertex 210. Lemma 3 holds on all 2,069 arcs. Of them, 2,046 have two distinct feet of odd valence (691 of valences three and five, 450 of valences three and three, 905 of valences five and five), 14 have a single foot of valence five, and 9 meet the vertex of valence two. Nearly every arc is thus a local dislocation or exchange. The twisted homology has dimension $518 = 2(n - 2)$, the pairing has rank 517, the dimension of the balanced lengths, and its kernel is the class across the connection; we conjecture that on a closed sphere without singular vertex of valence divisible by four the kernel is spanned by the classes across the connections. All 25 certificates of the conflicts examined have a representative, modulo the balances, whose odd part is a cycle of the graph of feet (exact), so Proposition 12 applies to all of them. The wrist certificate is, modulo the balances, the reading ℓ_{1902} of a comb whose two feet are the vertex 246: the winding separatrix. Configuration A is a closed walk of two combs between 76 and 78, the staircase $\ell_{611} + \ell_{624}$; C and D are walks of two combs, between 109 and 118 and between 155 and 156; B a walk of four combs through 102, 129, 103

and 105; the three turns at the right eye a walk of three through 76, 77 and 79. Configuration E is a closed walk of six combs through 198, 199, 204, 208, 207 and 209, of total index -2; F a walk of four combs through the vertices 14, 24, 23 and 15, all of valence three: a cap. The left hand is five times the dislocation ℓ_{1805} between 228 and 231 plus the tree of seven combs of Section 8.3; the set of seven of way 5 has 13 comb readings on 11 vertices, that of way 7 a tree of seven combs on eight vertices, and the set of fifteen 40 arcs, a component of 26 vertices of total index -8 and a walk of four combs through 0, 17, 18 and 20 (measured).

Question 2, read loosely, is true and empty. Allowing every local dislocation and exchange, that is every comb reading $2\ell_a$ and every closed walk of combs, Proposition 12 proves the conjecture for every certificate whose odd part is a cycle of the graph of feet, which is all those of the male body. But it follows from Farkas' lemma and a parity: nearly every arc is a comb, so every non-negative form on arcs whose feet are of odd valence is a sum of local dislocations, and the conjecture in this reading says nothing about which configurations of singularities conflict.

Read strictly, it is false. Restricted to staircases of weight one between vertices of odd valence, monotone closed walks round a cap and winding separatrices, as Question 2 states it, the certificates of configurations E and F are not non-negative combinations modulo the balances: exact rational balanced lengths make every such form non-negative and the certificate negative, -2.9967 for E and -1.9973 for F (measured, checked in integer arithmetic). These forms are infinitely many, but each is the length of a walk on a graph whose steps carry the arc lengths, so a negative one, if any, is a negative cycle or path, which a shortest-path search finds: the separation covers them all. The same test in floating point separates the set of four turns at the eyes (optimum -2, over 3,146 generators; not checked exactly) and finds the left hand inside the cone; the sets of seven and fifteen were not tested. The test is membership of the certificate in the cone of these local forms modulo the balances; that an extreme ray of the cone of certificates is neither a staircase nor a cap would not refute anything. Whether every certificate splits into combs on disjoint sets of vertices, each of the sign of the certificate, is open: the trees of the left hand and of way 7 share vertices between their combs.

Appendix D tests a lever before quantization suggested by Proposition 3, moving the singular vertices in the parametrization: in a map of fixed scale it barely moves the shifts.

8.5 Borders and handles

The bench holds models with borders (the eye sockets of Suzanne, the mouth of Jerry) and with handles (Bob, a torus; part and cage, of higher genus). This subsection gives the general form of the statements that need a closed sphere, and says which hold unchanged. A border node b is a corner of the flat metric of angle $a(b)\pi/2$; its *index* is $2 - a(b)$, zero on a straight border. For a border component β_j and lengths ℓ, let $L_j(\ell)$ be the sum of the lengths of its arcs.

Lemma 4 (Gauss–Bonnet with borders)**.** *Let D be a region bounded by curves of turns and by border components. Then*

$$\sum_{c\in D}(4-k(c)) + \sum_{b\in D\cap\partial\Sigma}(2-a(b)) = 4\chi(D) - \Theta,$$

where Θ is the turning of the curves of the turns at their corners, in quarter-turns: zero for the side of a simple curve.

Proof. Gauss–Bonnet for a flat surface with cone points and a piecewise straight boundary; border nodes are corners of the boundary, of turning $(2-a)\pi/2$. □

A side of a simple turn holding b border components without corner and h handles has index $4(1-b-2h)$: with a border it may hold no singular vertex at all. The lemma holds on the 1,076 sides of turns of Suzanne and Jerry; the sides that are not discs have total index 0 (one border) or -4 (two borders), 40 of them collars round an eye socket with no singular vertex; on Bob those holding the handle have -4, on part -4 and -12 (measured).

For positive lengths the homomorphism of the next theorem is the cycle parity of the quad graph, a $\mathbb{Z}/2$-homology invariant of any quadrangulation [31, Lemma 3.1] (see also [4]); the theorem extends it to the shifts and lengths of turns at all balanced lengths, borders included.

Theorem 1 (Parity of a turn on any surface)**.** *Let ℓ be whole and balanced. There is a homomorphism $\Omega_\ell\colon H_1(\Sigma;\mathbb{Z}/2)\to\mathbb{Z}/2$ such that for every turn*

$$\alpha_\gamma(\ell)+s_\gamma(\ell)\equiv\Omega_\ell([\bar\gamma])\pmod 2.$$

Its value on a border component is the parity of $L_j(\ell)$, the sum of all $L_j(\ell)$ is even, and for positive lengths $\Omega_\ell(z)$ is the parity of the length of any cycle of the quad graph in the class z; $\Omega_\ell=0$ exactly when the quad graph is bipartite. On a surface of genus zero with borders, $\alpha_\gamma+s_\gamma$ has the parity of the total length of the border components on either side of $\bar\gamma$, or of the number of times $\bar\gamma$ crosses any system of disjoint arcs joining the border components of odd length in pairs.

Proof. For a motion $M(x)=R^kx+t$ with t whole, the colour $p_1+p_2 \bmod 2$ of $M(x)$ is that of x plus $\pi(M)=t_1+t_2 \bmod 2$, so π is a homomorphism, and so is $\gamma\mapsto\pi(\hat h(\gamma))$. A loop round a singular vertex is a quarter-turn about a whole point, which keeps colours: it has $\pi=0$. The values being in a group of exponent two, the map factors through $H_1(\Sigma^\circ;\mathbb{Z}/2)$, and through $H_1(\Sigma;\mathbb{Z}/2)$ since the loops round the singular vertices span the kernel of $H_1(\Sigma^\circ)\to H_1(\Sigma)$. For a turn, $\hat h(\bar\gamma)$ is the translation $(\alpha_\gamma,\pm s_\gamma)$ up to a quarter-turn. Developed from one of its nodes, a whole point, a border component comes back displaced by $\sum_a\pm\ell_a v_a$ with v_a unit vectors of the axes, whose colour is that of $L_j(\ell)$. The border components add up to the boundary of Σ, zero in homology. Along a cycle of the quad graph every edge changes the colour. On a surface of genus zero the classes of the border components span $H_1(\Sigma;\mathbb{Z}/2)$ and $\bar\gamma$ is homologous to those on one of its sides; the last form is Lefschetz duality. □

On the head of Suzanne, whose eye sockets are borders of 23 and 25 edges, the turn round an eye socket is a ring of 23 quads at the lengths of one way and of 24 at those of another: Proposition 6 and Corollary 1(a) and (c) fail there. Its shift is zero for every balanced length, since a border without corner is a closed straight line of the flat metric:

Proposition 14 (Collars)**.** *Let the curve of a turn γ and a border component β_j without corner bound an annulus holding no singular vertex. Then $s_\gamma(\ell)=0$ and $\alpha_\gamma(\ell)=L_j(\ell)$ for all balanced lengths.*

Proof. A border without corner is straight, so its holonomy is the translation by L_j along its direction, an axis of the field. Cut along a path κ from $\bar\gamma$ to β_j, the annulus is a disc without cone point, whose border has trivial holonomy, as in Proposition 3(b): the translation of $\bar\gamma$ is that of β_j carried along κ, the vector L_ju with u one of $\pm e_1$, $\pm e_2$. For positive lengths its component along the walk, α_γ, is positive, so $u=\pm e_1$: the shift is zero and $\alpha_\gamma=L_j$; both identities are linear and extend to all balanced lengths. □

The theorem predicts the parity on all 660 turns of Suzanne, Jerry and a dome, 61 of them with $\alpha+s$ odd, and the proposition holds on all 40 collars (measured). On the output meshes, every closed loop has the parity the theorem gives: 36 of 36 on Suzanne, 5 of 5 on Jerry, 15 of 15 on a tube, 40 of 40 on a cylinder, 14 of 14 on Bob and 294 of 294 on the male body. In higher genus, the turns null in $H_1(\Sigma;\mathbb{Z}/2)$ all have $\alpha+s$ even (120 on Bob, 50 on part, 4 on cage, 30

on a cylinder), the parity is constant on homology classes and additive, and Ω_ℓ depends on the lengths: the three ways of Bob give it the values $(0,1)$, $(0,0)$ and $(1,0)$ on two handle loops, and 22 of the 80 handle loops of one of its meshes have odd length (measured).

Proposition 15 (Burgers sum with borders and handles)**.** *Let the loops λ_i round the singular vertices, β_j round the border components and a_k, b_k round the handles generate the fundamental group of Σ°. For every word w in them,*

$$\tau(w) = \sum_i \rho\Big(\frac{\partial w}{\partial x_i}\Big)(I - R_i)\, p_i + \sum_{g \in \{\beta_j, a_k, b_k\}} \rho\Big(\frac{\partial w}{\partial g}\Big)\tau(g).$$

For a turn, the shift is the Burgers sum of the singular vertices plus a period term*, linear in the lengths and not determined by the developed positions of the singular vertices. A border component without corner has holonomy the translation by L_j along its developed direction; a handle inside the region between two turns contributes the commutator $(I - \rho(b))\tau(a) - (I - \rho(a))\tau(b)$, zero when both its loops have trivial rotation.*

Proof. Both sides are maps ϕ with $\phi(uv) = \phi(u) + \rho(u)\phi(v)$ that agree on the generators, as in the proof of Proposition 10. The commutator of $x \mapsto Ax + a$ and $x \mapsto Bx + b$ with commuting rotations is the translation by $(I - B)a - (I - A)b$. □

With its border term, the sum reproduces the translation of all 314 sides holding a border on which we tested it; without it, it reproduces none of the 586 such sides of Suzanne and Jerry (measured). Sides that are discs follow Proposition 3(a) verbatim (86 of 86 on Bob, 25 of 25 on part); on a torus without singular vertex the shifts are $\pm\ell_0$ or $\pm\ell_2$, pure periods (measured). With the period term, Remark 4, the staircases, Theorem 2 and Corollary 3, which use no topology, hold on any surface; when the twisted windings of the handle loops vanish, a combination of shifts is again a form in the developed positions of the singular vertices alone.

Proposition 16 (Two crossing turns on a torus)**.** *On a closed surface of genus one, let the curves of two turns be simple and cross once. Cut along both, the surface is a disc D holding every singular vertex, and its Burgers sum vanishes: $\sum_i Q_i(I - R_i)p_i = 0$, in the order of first visit, for all balanced lengths. More generally, if κ is a simple loop crossing the curve of a turn once, with rotation R^j, then $(I - R^j)\, T_\gamma = \pm B_D$: for $j \neq 0$ the translation of the turn is set by the singular vertices, for $j = 0$ it is a free period and the singular vertices satisfy $B_D = 0$.*

Proof. The border of D is the commutator of the two curves, whose holonomies are translations; their commutator is the identity, and Proposition 3(a) applies to the disc D. With κ of rotation R^j, the commutator is the translation by $(I - R^j)T_\gamma$. □

On the T-mesh of Bob, of genus one with ten vertices of valence three and ten of valence five, the Burgers sum of the twenty vertices vanishes in all 304 pairs of simple turns crossing once, and in none of 90 trials with the order shuffled (measured). With a single pair of valences three and five the sum is $(I - R^{\pm 1})(p_5 - p_3)$: a torus with two turns crossing once must develop the two cone points onto one point. That no such torus exists at all is the theorem of Barnette, Jucovič and Trenkler [6]; it also follows from Abel's theorem, the quartic differential $(dz)^4$ of the flat metric having the divisor $c_5 - c_3$, which is principal on an elliptic curve only if $c_5 = c_3$. The obstruction is to existence, not to closing. Proposition 16 with $j = 0$ is the commutator argument discussed by Izmestiev et al. [29, §7]: it excludes a 3,5-quadrangulation of the torus with two turns crossing once, but not one whose handle loops rotate, which their Holonomy Theorem handles. The conflict actually met on Bob is of another kind: two turns that wind three and two times round the same handle, whose shifts carry its period with the coefficients 3 and 2; the combination $3s_{33} - 2s_{31}$ cancels the period and equals, modulo the balances, the sum of two arc lengths at a vertex of valence three, one of them held open (measured). On Bob, each of the 102 held turns closes on its own, and all of them close together in five of the six ways (measured).

What holds unchanged. Propositions 1 and 2 (with open strips at the borders), the remarks on whole periods and whole solutions, Lemma 1 for discs and annuli as stated, Propositions 3, 4, 5 (for annuli), 7, 8, 9 and 11, Remark 4, Theorem 2 and Corollary 3 need no closed sphere. Lemma 2, Proposition 6, Corollary 1, the count of independent rings, the setting of Proposition 10 and Corollary 2 do, and are replaced by the statements above; Questions 1 and 2 were stated for a closed sphere. Appendix E carries the note to the rotations of order six and three of triangle and hexagonal meshes.

9 Turns every quantization within bounds leaves shifted

The obstructions of Section 8 say that a set of turns cannot close together. Turned around, they bound from below the number of turns of a given family that any quantization in a set P of lengths leaves shifted, and so tell how far a heuristic such as the one of Section 4 is from the best possible within P. On the male body, P is defined by the engine's bounds of separation (Section 7), so the measured bounds are relative to them, not to the singularity configuration alone. We are not aware of an earlier lower bound of this kind for helices (Section 2).

Theorem 2 (Forced shifts)**.** *Let H be a finite set of turns of a T-mesh and P a non-empty set of lengths containing every quantization considered: for instance the balanced lengths meeting given bounds of separation, $\ell \geq b$, possibly with a cap $\frac{1}{n}\sum_a |\ell_a - t_a| \leq D$ on the mean deviation from the targets t, or with a given number of quads. Let*

$$\tau_P(H) = \min\{\, |X| : X \subseteq H, \text{ the turns of } H \setminus X \text{ can be closed together by real lengths in } P \,\}.$$

Then for every $\ell \in P$, whole or not, at least $\tau_P(H)$ turns of H have $s_\gamma(\ell) \neq 0$. If H contains k pairwise disjoint subsets none of which can be closed within P, then $\tau_P(H) \geq k$.

Proof. At a whole $\ell \in P$, the turns of H with zero shift are closed by ℓ itself, which is real and lies in P; so the shifted turns form a set X as in the definition, and $|X| \geq \tau_P(H)$. A set X whose removal leaves a closable set meets each of the k subsets, which are disjoint, so $|X| \geq k$. □

The theorem is elementary; its content is in what makes subsets infeasible, which Section 8 relates to the singularities. The proof uses neither convexity, nor linearity of the shifts, nor whole lengths; convexity is what lets $\tau_P(H)$ be bounded by linear programs, as below. In the language of linear feasibility, the minimal non-closable sets are irreducible infeasible subsystems (IIS) and $\tau_P(H)$ is a minimum IIS cover [3, 49]; the theorem is the packing bound of that covering problem, applied to the shifts of turns. The bounds of separation are what make the shifts of a dislocation non-zero, and the statement becomes one about parallel rings.

Corollary 3 (Chains of dislocations)**.** *Let $\gamma_0, \gamma_1, \ldots, \gamma_m$ be turns and suppose that for $i = 1, \ldots, m$*

$$s_{\gamma_{i-1}}(\ell) - s_{\gamma_i}(\ell) = \varepsilon_i\, F_i(\ell) \quad \text{for all balanced } \ell,$$

with $\varepsilon_i \in \{+1, -1\}$ and forms F_i that are positive on the whole points of P; for instance F_i the length of a staircase (Proposition 7) between a vertex of valence three and one of valence five lying between γ_{i-1} and γ_i (Propositions 4 and 3(b)), the staircase containing an arc held at length at least one. Let q be the number of indices $i < m$ with $\varepsilon_{i+1} \neq \varepsilon_i$. Then for every whole $\ell \in P$ at most $q + 1$ of the turns γ_i are rings: at least $m - q$ are helices. In particular $m + 1$ nested parallel turns separated by m dislocations whose staircases all climb the same way leave at least m helices.

Proof. If γ_i and γ_j, $i < j$, are both rings, then $\sum_{k=i+1}^{j} \varepsilon_k F_k(\ell) = s_{\gamma_i}(\ell) - s_{\gamma_j}(\ell) = 0$; the $F_k(\ell)$ being positive, the signs $\varepsilon_{i+1}, \ldots, \varepsilon_j$ are not all equal, so a change of sign lies between i and j. Between two consecutive rings of the chain lies a change of sign of its own, so there are at most $q + 1$ rings. (Equivalently, in Theorem 2, every pair of rings of a monotone stretch is an infeasible pair.) □

Several chains, or disjoint conflicts of any of the kinds of Section 8.2, add up by the second part of the theorem. The corollary is stated with signs ε_i and forms F_i that the singular vertices between the turns determine: by Proposition 3(b), $s_{\gamma_{i-1}} - s_{\gamma_i}$ is the across component of the Burgers vector of the annulus between them, computed from the positions and valences of its singular vertices alone.

Remark 7 (What the bound counts)**.** *The theorem counts turns with a non-zero shift. At the lengths where a turn is a strip of the mesh (Proposition 2) a non-zero shift is a helix; at other lengths the same walk may not be a strip. One spiral strip may also carry several turns of H, and the measure of Section 5 counts edge loops that spiral within four quads, which is yet another count. The bound is therefore a bound on turns, and is compared below with the turns the engine leaves shifted.*

The bound is a transversal number. The least number of turns left shifted is a covering number of the minimal conflicting sets, and it is attained. This is the standard correspondence between minimum IIS covers and maximum feasible subsystems [3, 49], stated here for turns.

Theorem 3 (Forced shifts as a transversal number; standard)**.** *Call $Y \subseteq H$* closable *in P if some $\ell \in P$ has $s_\gamma(\ell) = 0$ for all $\gamma \in Y$, and a* core *a set that is not closable while all its proper subsets are. For any P: (i) $H \setminus X$ is closable if and only if X meets every core; hence $\tau_P(H)$ is the least size of a set meeting every core; (ii) for every $\ell \in P$ the turns ℓ leaves shifted meet every core; (iii) every least such set is exactly the set of turns left shifted by some $\ell \in P$; (iv) $P \subseteq P'$ implies $\tau_P(H) \geq \tau_{P'}(H)$.*

Proof. Closability is hereditary. If X misses a core, $H \setminus X$ contains it; if $H \setminus X$ is not closable it contains a minimal non-closable subset, a core, which X misses. The zero-shift set of ℓ is closed by ℓ, so its complement meets every core. For a least X, a length closing $H \setminus X$ shifts a subset of X meeting every core, hence all of X. Part (iv) is immediate. □

So $\tau_P(H)$ lies between the largest number of disjoint cores, the lower bound of Theorem 2, and the least hitting set of the cores found so far, which is exact as soon as its complement is closable: an *implicit hitting-set* loop alternates the two. It is the algorithm of Parker and Ryan [49], an instance of the implicit hitting-set approach [46]; we find cores by the deletion filter of Chinneck and Dravnieks [16].

Proposition 17 (Whole lengths and size without a cap)**.** *If P is given by the balances and bounds $\ell \geq b \geq 0$, a set closable in real lengths is closable in whole lengths, and closable in real lengths at any number of quads greater than $\inf_{P_Y} Q$, and at $\inf_{P_Y} Q$ itself when the infimum is attained, where P_Y is the set of lengths of P closing Y and $Q(\ell)$ the number of quads.*

Proof. Scale a rational point of P_Y by its denominator (Remark 5). For a number of quads q, take $\ell \in P_Y$ with $0 < Q(\ell) \leq q$; for $c \geq 1$, $c\ell \in P_Y$ and $Q(c\ell) = c^2 Q(\ell)$, a continuous function of c that reaches q. □

With a cap on the deviation, P_Y is a polytope and $Q(P_Y)$ an interval, since P_Y is convex and Q continuous: two exact points of P_Y, one on each side of a window of sizes, certify that Y closes within it. Certifying that it does not needs the extremes of Q over a polytope, an indefinite bilinear program, for which the McCormick relaxation was of no use here. The cap acts as a budget shared by all the turns: with $D = 0.474$ the lengths may move by $\sum_a |\ell_a - t_a| \leq 980.7$ edges in all, while closing the 109 strips of way 2 that the bounds alone do not force costs 1,680.2 (the balances alone 176.4). The cores then become covers of a knapsack, of 1, 50 and 49 turns for the first three disjoint ones, and the savings of closing several turns are not additive (measured).

Measured on the male body. For each way of Section 7 we took for H its held turns, each up to sign once, and for P the balanced lengths meeting the bounds of separation; then also the cap $D = 0.474$ on the mean deviation, that of the engine version that made the export (1.5 times the 0.316 of its plain search; the engine at the cited commit allows 0.516), with the targets of that way; then also whole lengths and the engine's window of sizes, from 10,379 to 10,801 quads, with Q counted exactly as the sum over the patches of the products of their sides. We also took for H the strips of the engine's own mesh for way 2 (the way it keeps) whose real shift leans less than 0.12, the rings of Appendix F: 222 ways round, 111 strips. With the bounds alone the implicit hitting-set loop computes $\tau_P(H)$ exactly in every family of held turns, in at most 66 seconds of wall-clock time each (between 0 and 66; hardware in Section 6), with cores found by a deletion filter and certified by HiGHS. With the cap, upper bounds come from closable subsets found by a mixed-integer program, checked feasible, and lower bounds from disjoint cores and the solver's proven bound; with the window and whole lengths, upper bounds are whole points checked in exact arithmetic, and lower bounds those of the real relaxation and the solver (Table 6).

way	turns	bounds	bounds, cap	cap, size, whole	left shifted by the engine
0	48	1	1	1	13
1	78	5	5	5–6	39
2	52	1	1	1	16
3	86	4	4	5–18	44
4	84	1	1	1–12	49
5	160	6	7–17	7–86	115
6	88	1	1	1–17	44
7	169	9	9–26	9–90	123
2, strips of the mesh	111	2	3–11	3–43	81

Table 6: Male body at 10,590 quads: turns of each family that every quantization in P leaves shifted (Theorems 2 and 3), with P given by the engine's bounds of separation (Section 7), then also by the engine's cap on the mean deviation (real lengths), then also by whole lengths and the engine's window of sizes. A single number is exact; a range is certified: the solver's proven bound or the disjoint cores below, a closable subset found within the time limit (120 to 1,800 seconds) above. Last column: turns of the family with a non-zero shift at the engine's whole lengths for that way. Turns are counted up to sign; "strips": the ways round of the engine's mesh for way 2 leaning less than 0.12, two per strip.

With the engine's bounds alone, the forced turns are few and exactly known: 1, 5, 1, 4, 1, 6, 1 and 9 held turns in ways 0 to 7, and 2 strips. In ways 0, 2, 4 and 6 the one forced turn is the turn round the left wrist of Section 8.2; in the other ways the cores are the irreducible sets of Table 4 and their like, and in way 7 nine turns are needed although no more than eight of its 21 cores are disjoint, so the packing bound of Theorem 2 is not always attained. By Theorem 3(iii) each value is the number of turns left shifted by some lengths in P. Of the 111 strips of the engine's own mesh that lean like rings, 2 must stay shifted within P, one of them round the wrist; the engine leaves 81 shifted, and the measure of Section 5 counts on that mesh 1.1% of spiral edges (a single run of the engine version that made the export; later versions give 1.15% to 1.59%, in 8 to 9 spiral loops).

With the engine's cap and its window, at whole lengths, the exact value is still 1 in ways 0 and 2, against 13 and 16 left shifted by the engine: whole lengths closing all held turns but the wrist exist at 10,439 and 10,436 quads, at mean deviations 0.442 and 0.469 (measured, exact). Elsewhere the cap acts as a shared budget, the cores grow to dozens of turns, and the intervals are wide. In way 3 the real relaxation is not exact: real lengths close all but 4 held turns at the right size, whole lengths within the cap leave at least 5 (solver). The engine's own lengths of

way 3 make 10,810 quads, outside the window. Two caveats: the cap we impose is that of the engine only approximately (a quantization at 0.4737 was refused by it, and the meshes below use 0.470); and the bounds of separation are those the integer search placed at the engine's lengths, so optima that bring other arcs to zero can break the mesh (one made 9,426 quads with 90 inverted faces). Keeping open every arc the engine keeps open gives a *sound* subset of P, and upper bounds.

Where the shifted turns come from. For the held turns of way 2 the gap between the engine and the optimum splits exactly: of the 16 turns the engine leaves shifted, its bounds of separation force 1, the cap 0 more, the size 0, whole lengths 0, and the heuristic 15; in the sound subset the optimum is 4, the heuristic's share 12. In the other ways the heuristic leaves at least 12, 33, 26, 37, 29, 27 and 33 turns more than the best lengths found (ways 0, 1 and 3 to 7). On the strips of the mesh the engine keeps, the bounds of separation force 2 shifted strips, the cap between 1 and 9 more in real lengths, the size at most 2 more (not certified); integrality is not resolved (between 0 and 32 more). A local search over whole lengths within the same cap and size closes 38 more strips than the engine (43 shifted, at a deviation of 0.473 and 10,495 quads), and 15 more at the engine's own deviation of 0.328; within the sound subset, 54 and 69 shifted: most of the gap is the heuristic. Closing them does not reduce the spirals, however. At the new lengths the strips are other walks (234 ways round, 162 helices, against 230 and 168), and the spiral share counted on the strips is 0.020 against 0.011 for the engine, on a mesh that the unsound lengths break; the sound closing of 54 gives 0.050, and four optimal closings of the held turns of way 2 give 0.003, 0.058, 0.005 and 0.027, and, meshed by the engine, two of them 0.52% and 28.1% of spiral edges (measured). The number of turns closed is thus not what the measure of spirals sees: the strips move with the lengths (Appendix F).

9.1 Choosing the turns

The heuristic leaves most of the gap. One may hope for an exact method fast enough for the engine; in general there is none, unless P = NP. Call Ring-Closing the problem: given the balances B, bounds b, a set H of linear forms (the shifts) and a number k, are there lengths in $P = \{\ell : B\ell = 0,\ \ell \geq b\}$ that close at least k forms of H? Choosing the most equations of a linear system that hold together, some constraints being mandatory, is the maximum feasible subsystem problem, NP-hard and hard to approximate in general [2]; minimum covers of irreducible infeasible subsystems are strongly NP-hard already for network flow systems [30]. The theorem shows that the hardness survives the special structure of quantization: balances of a bidirected graph, bounds 0 and 1, forms with coefficients 0 and 1; the general case is known, this restricted case is new as far as we found.

Theorem 4 (Choosing the turns is NP-hard)**.** Ring-Closing *is NP-complete, in real or in whole lengths, already when B is the incidence matrix of a bidirected graph, the bounds are* 0 *or* 1*, and the forms have coefficients* 0 *or* 1*; it is strongly NP-hard, and stays NP-hard with a cap on the deviation, which may be loose. Moreover $\tau_P(H)$ can equal the vertex cover number of any graph, with $|H|$ its number of vertices. So it cannot be approximated within $\sqrt{2} - \varepsilon$ unless $P = NP$ [19, 20, 35, 36], nor within $2 - \varepsilon$ if the unique games conjecture holds [34], and the largest closable subset of H cannot be approximated within $|H|^{1-\varepsilon}$ unless $P = NP$ [25, 62].*

Proof. Membership: given a set Y of k forms, closing them in P is the feasibility of a linear program, and by Proposition 17 real and whole closability agree. Hardness, by reduction from Independent Set: for a graph G, take for each edge $e = uv$ four arcs $y_{u,e}$, $y_{v,e}$, d_e, d'_e with the three balances

$$y_{u,e} + y_{v,e} - d_e = 0, \qquad y_{u,e} + y_{v,e} - d'_e = 0, \qquad d_e - d'_e = 0,$$

in which every arc appears in exactly two rows with coefficients ± 1, the bounds $d_e, d'_e \geq 1$ and $y \geq 0$, and for each vertex u the form $s_u = \sum_{e \ni u} y_{u,e}$. With $y \geq 0$, $s_u = 0$ forces $y_{u,e} = 0$ on every edge at u. If two adjacent vertices u, v are closed, the edge uv gets $d_e = y_{u,e} + y_{v,e} = 0 < 1$. Conversely, for an independent set Y, every edge has at most one end in Y: put $y = 0$ on the side of that end and 1 on the other side, and on an edge with no end in Y put $y_{u,e} = 1$ and $y_{v,e} = 0$ (either way); then $d_e = d'_e = 1$ on every edge, and these are whole lengths in P closing Y. So the closable sets are exactly the independent sets of G, the largest has the size of the independence number, and $\tau_P(H) = |V| - \alpha(G)$ is the vertex cover number. The instance has size linear in G and numbers 0 and ± 1. A cap large enough constrains nothing. □

We checked the reduction by computer on 40 random graphs, comparing the closable sets with the independent sets (measured). Its third balance is redundant over the reals; it is there so that every arc lies in two balances, as in a T-mesh. The best known hardness of approximating vertex cover without a conjecture was 1.3606 [18], before $\sqrt{2} - \varepsilon$. The theorem is about the abstract problem: whether the instances that T-meshes produce are as hard is open.

Conjecture 1 (Realizable instances)**.** Ring-Closing *stays NP-hard when B, b and H are the balances, the bounds of separation and the turns of a T-mesh of a closed surface of genus zero.*

The evidence is that vertex cover stays NP-hard on planar graphs of small degree [24], and that two turns separated by a dislocation conflict like the two ends of an edge (Proposition 4).

What is easy. Several parts of the problem are polynomial. Closing a given set Y is a linear program. Without turns and without a window, quantization in whole lengths is an integer minimum-cost flow on a bidirected graph, a matching problem in the sense of Edmonds and Johnson [22], which is polynomial; the exact refinement of [26] works in this setting. When the cores are known and disjoint, $\tau_P(H)$ is their number; when every core has two turns and the graph of conflicts is bipartite, it is a maximum matching (Kőnig). When the shift forms span, modulo the balances, a space of fixed dimension d, the closable sets are the flats of an arrangement and can be listed in time polynomial for fixed d; but d is at most twice the number of singular vertices, and on the male body the held turns of a way have rank 52 of 52 and the strips 107 of 111, so this does not help. The implicit hitting-set loop above is exact and fast when the cores are small, as with the bounds alone (66 seconds at most per family); with the cap the problem is a budgeted maximum feasible subsystem with a lattice on top, and the cores grow.

Remark 8 (The window of sizes; a sketch)**.** *The window on the number of quads makes even closing nothing hard in the weak sense. Take a disjoint union of tori, each of two patches whose sides have the targets Nv_i and 1, so that $Q = \sum_i 2a_i c_i$, and a cap that lets the short sides c_i grow by T_0 edges in all. For N large, the window $Q = 2N(\sum_i v_i + T_0)$ is met exactly when $\sum_i v_i(c_i - 1) = T_0$ with whole $c_i \geq 1$: unbounded subset sum, which is NP-complete [44], with numbers exponential in the size of the instance (the gadgets are of genus one and disconnected). Whether the window alone is strongly NP-hard, with Q polynomial in the size, is open, as is the complexity of maximizing Q over a polytope of lengths in reals.*

An exact selection, end to end. Appendix G measures what an exact selection buys on the bench: a mixed-integer program that closes as many strips as it can, within the engine's cap and budget, gives lengths that do not spiral less than the engine's, but good extra candidates for the engine to choose among (from 4.54% to 2.32% and 2.70% of spiral edges over 36 runs), and only when the solver finishes: with a limit of 5 seconds per solve the gain vanishes.

Status of Sections 8 to 9. *Proved:* Lemmas 1, 2, 3, 4 and 5; Propositions 3 (with the order of first visit made precise), 4, 5, 6, 7, 8, 9, 10, 11, 13 (on a closed sphere without vertex of valence divisible by four), 12 (the reading of closed walks as twisted cycles is only a remark), 14, 15, 16, 18 and 17, Corollary 2, Theorems 1, 2, 3 and 4, and Corollary 3, each under its hypotheses; Remark 4, and the existence of whole solutions when real ones exist. *Machine-checked:* the algebraic and combinatorial cores listed in Section 10; the other statements, listed there too, are proved on paper only. *Measured:* on the T-mesh of the male body, every statement of Section 8.1 wherever it applies (Appendix B); the configurations of the conflicts (Table 5), the staircases, the separatrix at the wrist and the windings of the larger sets, as identities modulo the balances, without the homotopies; the Fox–Burgers formula on all 1,299 turns and the checkerboard rule for combinations on all of them and on 14 infeasible sets; the combs of all 2,069 arcs, the rank of the Burgers pairing and the comb readings of all 25 certificates; the Burgers sum on 3,600 of 3,604 disc-bounding rings of seven other models, and the response of the shift to moving a singular vertex or rotating the field (Appendix D); the parity, collars, border and period terms on Suzanne, Jerry, Bob, part, cage and simple shapes (Section 8.5); the four turns whose winding numbers are related while their translations are not; the dislocation of weight 5 at the left hand; the structure and ranks of the eight systems, the 171 implicit zeros and their certificates, the bounds 0.817, 0.758 and 0.536, the infeasible sets of Table 4 (exactly where marked), the forced shifts of Table 6, the end-to-end selection and the meshes. *Refuted:* that the conflicts between two turns are pairs of valences three and five on one trace (0 of 13); Question 1 as stated; Question 2 read strictly (configurations E and F, exactly); a formula for combinations of shifts in the winding numbers alone (in principle and on the body); the parity of Proposition 6 beyond the closed sphere (Suzanne). *Conjectured:* that the kernel of the Burgers pairing is spanned by the connections; that the readings of closed walks of combs are Burgers forms of twisted cycles; that Ring-Closing stays NP-hard on the instances of T-meshes (Conjecture 1); a rule modulo three with borders on meshes of order three; that moving singularities in a map of fixed scale cannot set the shifts (Appendix D), a reading of measurements. Question 2 read loosely is proved (Proposition 12) and empty.

10 What is machine-checked

The folder `docs/research/closing-helices/lean` of the repository is a Lean 4 [17] project on Mathlib [54], pinned to Mathlib `v4.35.0` and the toolchain `leanprover/lean4:v4.35.0`; `make proofs` fetches Mathlib compiled and checks it. It holds 187 theorem declarations, many of them small lemmas, a few of them tautologies that only restate a hypothesis (such as `dislocation_closes_iff` or `staircase_rings`); the count is not evidence, and the folder's `README.md` maps each statement of the note to its main Lean theorems. Every theorem depends only on Lean's standard axioms. What Lean checks is the algebraic and combinatorial core of each statement below, with the topological facts it uses (that a loop develops into a given motion, that a curve bounds a disc, that the fundamental group of Σ° is presented by the loops λ_i, that the quad graph of a sphere is bipartite) as explicit hypotheses of the theorem that uses them.

- Checked as algebra: the Burgers sum and the order of first visit (Proposition 3), dislocations, staircases, exchanges and the conjugation step of Proposition 9; the checkerboard rules, the induction of the proof of Proposition 6 given a 2-colouring, and the second proof of Remark 2; the Fox–Burgers formula, its corollary and the commutator; the handshake on a closed surface, Proposition 18 (stronger than stated), the Farkas certificates of Remarks 4 and 5, Theorem 2 and Corollary 3; for fields of any order, the rotation part of Gauss–Bonnet, the Gauss–Bonnet counts of triangle and hexagonal meshes, the lengths of dislocations and the congruences of Section 8.5.

- Checked in a weaker form than the statement: Proposition 1, only the additivity of the composed translation given additive crossings (the integer coefficients and the reading as a helix are not formalized); Lemmas 1 and 2, only the Euler-characteristic count they use (which corners are convex, that the regions are discs, and the case analysis of the crossing are hypotheses); Proposition 2, only in a model where the strips are *defined* as the cycles of a successor map built from a permutation of the half-integer stations, so that the bijection holds by construction, together with the fact that the real walk from $k+\frac{1}{2}$ follows the unit edges in a one-dimensional model of the gluing; that the station walk of the gluing data is a permutation (a closed surface, an injective gluing), that every patch side is positive, and that the model is the quad mesh, are assumptions.

- Proved on paper only, without a human or Lean check: Propositions 13, 12, 14, 15, 16 and 17, Lemmas 3 and 4, Theorems 1, 3 and 4, the case of Lemma 5 with borders, and the statements on rotations of order six other than those above. An independent referee pass found errors among these (Propositions 13 and 12, Theorem 4), and in two statements whose Lean check is weaker than the statement (Propositions 1 and 2); all are corrected here.

Formalizing found no error of sign or convention; it sharpened the hypotheses of Propositions 6, 18, 2 and 7 and of Remark 4, which are stated here in their sharpened form (the `README.md` lists the changes). Appendix H lists what is checked statement by statement.

11 Open problems

Open problem 1 (Exact formulation)**.** *Minimize the deviation of the quantization subject to balance, validity and $s_\gamma(\ell) = 0$ for a set of turns. This is a minimum-deviation flow with linear side constraints; how much of the exact machinery of [26] carries over is open. Section 7 shows that the linear relaxation already closes almost all held turns; with the number of turns closed as the objective, the problem is NP-hard (Theorem 4).*

Open problem 2 (Closing in whole numbers)**.** *The solutions of the balance and shift equations in whole numbers form a lattice, and Appendix F searched it through a reduced basis and the reformulation of [1]: whole lengths closing every ring asked for are found, but in minutes, with the number of quads held only to first order, and at a deviation* 0.14 *to* 0.22 *above the real optimum. Can the integer search of the quantizer move within this lattice at the speed of a flow solver, with the number of quads held exactly? This matters once the right turns are chosen; Appendix F suggests it is not, on its own, where the loops are lost.*

Open problem 3 (Which turns)**.** *This is, on the evidence of Appendix F, the central problem: closing rings changes the lengths, the lengths change the strips, and the strips of the new mesh are other walks. The straight walks that return are infinite in number; we sample them, and Appendix F shows the sample misses most of the loops that spiral. Choosing the most turns to close is NP-hard (Theorem 4), and closing the most rings, even exactly and keeping each ring the same walk, gives good candidates only when the solver finishes, and does not by itself make fewer spirals (Section 9.1, Appendix G). Is there a finite set of turns, a basis in some homological sense of the regular region of the T-mesh, whose closing implies that every loop of the quad mesh closes? Can the loops of the quad mesh be read off the T-mesh and its lengths directly, so that every one is accounted for? Which objective, short of all loops, predicts the spirals the measure sees?*

Open problem 4 (Feasibility)**.** *On a human torso, some held turns cannot be closed together: the projection lets them go. Our hypothesis was that pairs of singularities between two rings force a dislocation, and that closing all rings needs the poles to face one another, which the*

separation of singularities forbids at the scale of the T-mesh; Propositions 4 and 7 prove this for a single pair joined by a straight chain or a staircase. A capped cylinder shows the same conflict: its layout of six patches exists, but only with arcs our quantizer will not bring to zero. On the male body the conflict comes down to one arc in four of the eight ways, to 9 to 30 arcs in the others (Section 7), and to irreducible sets of one to fifteen turns; the smallest are dislocations, exchanges of two vertices, caps and a winding separatrix (Section 8.2). When is a set of turns simultaneously closable, in terms of the singularity configuration? Every certificate is a sum of combs (Proposition 12), which says nothing about the singularities; restricted to staircases, caps and winding separatrices it is not (Section 8.4). A useful answer must lie between: for instance a decomposition into combs on disjoint sets of vertices, each of the sign of the certificate.

Open problem 5 (Lower bounds with the size)**.** *Theorem 2 bounds the shifted turns any quantization leaves within a set of lengths, and on the male body the engine's bounds of separation force exactly one to nine held turns. With the cap and the window of sizes the values are known only within wide intervals (Table 6): a closable set at a size is certified by two exact points, but its absence needs the extremes of the number of quads, an indefinite quadratic function, over a polytope. How can $\tau_P(H)$ be computed for quantizations of a given size and deviation?*

Open problem 6 (Complexity)**.** *Theorem 4 is about abstract instances. Is* Ring-Closing *still NP-hard on the instances of T-meshes of a closed sphere (Conjecture 1), or for a fixed number of singular vertices? Is the window of sizes alone strongly NP-hard, with a number of quads polynomial in the size of the instance, and what is the complexity of the largest number of quads over a polytope of lengths? And when the set of turns depends on the lengths, as the strips of the mesh do, what is the problem at all?*

Open problem 7 (Finite trajectories)**.** *In the language of [40]: given a singularity configuration on a closed surface of genus zero, when does a quartic differential with all trajectories finite exist, and how close to a given cross field can it be chosen? Our turns are its discrete, integer version.*

Towards proofs. The algebraic and combinatorial cores of Sections 3 and 8 to 9 are machine-checked (Section 10); their topological hypotheses are not, nor the results on combs, borders and complexity. Proposition 1 is elementary but its statement over a T-mesh with T-junctions is easy to get wrong. A formal model of an abstract complex of rectangles with integer side lengths, its turns and its developing map would turn those hypotheses into theorems, and could carry the sufficiency question of the third problem for special classes of layouts.

Reproducibility. The method is implemented in openclare (Rust), in the main branch of the repository at the cited commit; the bench is driven by `bench/run.py` and the marks by `scripts/scores.py`. The export of the T-mesh used in Sections 7 to 9 and the appendices is an experimental hook of the branch `rd-exact`, and the end-to-end selection of Appendix G is on the experimental branch `maxclose-rd`; neither is part of the engine, and neither is tagged yet. The scripts and logs of the exact experiments are not in the repository (Appendix A): until they are published, the measured statements of those sections can be checked against the branches but not rerun as they stand. The Lean proofs are in `docs/research/closing-helices/lean` and replay with `make proofs`. This version of the note was built from commit `3b890ecc8080`.

AI assistance

This note and the software it describes were made by a human author working with an AI system. We state how the work was divided. No AI system is an author of this note.

The human author chose the problem and its aims: edge loops that close into rings, the quad sphere with the eight poles of a cube as a target, and, as an artist, a mesh with few singular vertices and few spirals rather than perfectly square quads, always made of quads only. He set the direction of the research and the order of the work, decided which results to pursue, which changes to merge into the engine and which to drop, authorized the tools (LaTeX, Lean and Mathlib), and judged the results through the measurements, pictures and drafts presented to him.

The AI system , Claude Opus 5.5 (Anthropic), used through Claude Code in October 2026, with several sub-agents of the same model working in parallel, did the rest. It wrote the changes to the engine and the experiment scripts, ran all the experiments, the bench, the solver runs and the exports of the T-mesh, carried out the exact computations (SymPy, HiGHS through SciPy), searched the literature, stated and proved the propositions, wrote the Lean formalization and drafted the text of this note. References were checked for existence, and the note attached to each entry of the bibliography says how much of the work was read: the abstract only, a part, or all of it. The novelty of the results was assessed by AI agents, for several of the closest works from their abstracts only.

What was checked, and how. The algebraic and combinatorial cores listed in Section 10 are machine-checked in Lean, using only Lean's standard axioms. The topological statements that Lean takes as hypotheses, and the statements proved on paper only (listed in Section 10), were checked by AI agents rereading them and testing them numerically on the data, not by a human expert; three AI review passes (bibliography, prior art, a referee report) preceded this version and found errors among the paper-only proofs, now corrected. Appendix A says where the raw output of each experiment is, and the paragraph "Reproducibility" what can be rerun today. The note has not been peer reviewed. Errors remain possible, and the human author, who publishes the note, is responsible for its content.

References

[1] Karen Aardal, Cor A. J. Hurkens, and Arjen K. Lenstra. Solving a system of linear Diophantine equations with lower and upper bounds on the variables. *Mathematics of Operations Research*, 25(3):427–442, 2000. doi: 10.1287/moor.25.3.427.12219. Abstract read.

[2] Edoardo Amaldi and Viggo Kann. The complexity and approximability of finding maximum feasible subsystems of linear relations. *Theoretical Computer Science*, 147(1–2):181–210, 1995. Abstract read.

[3] Edoardo Amaldi, Marc E. Pfetsch, and Leslie E. Trotter, Jr. On the maximum feasible subsystem problem, IISs and IIS-hypergraphs. *Mathematical Programming*, 95(3):533–554, 2003. doi: 10.1007/s10107-002-0363-5. Abstract read.

[4] Dan Archdeacon, Joan Hutchinson, Atsuhiro Nakamoto, Seiya Negami, and Katsuhiro Ota. Chromatic numbers of quadrangulations on closed surfaces. *Journal of Graph Theory*, 37 (2):100–114, 2001. doi: 10.1002/jgt.1005. Abstract read.

[5] László Babai. On Lovász' lattice reduction and the nearest lattice point problem. *Combinatorica*, 6(1):1–13, 1986. doi: 10.1007/BF02579403. Abstract read.

[6] David Barnette, Ernest Jucovič, and Marián Trenkler. Toroidal maps with prescribed types of vertices and faces. *Mathematika*, 18(1):82–90, 1971. doi: 10.1112/S0025579300008408. As cited by Izmestiev et al.; not read.

[7] David Bommes, Henrik Zimmer, and Leif Kobbelt. Mixed-integer quadrangulation. *ACM Transactions on Graphics*, 28(3):77:1–77:10, 2009. doi: 10.1145/1531326.1531383. Proc. SIGGRAPH 2009.

[8] David Bommes, Timm Lempfer, and Leif Kobbelt. Global structure optimization of quadrilateral meshes. *Computer Graphics Forum*, 30(2):375–384, 2011. doi: 10.1111/j.1467-8659.2011.01868.x. Proc. Eurographics 2011. Read in full.

[9] David Bommes, Marcel Campen, Hans-Christian Ebke, Pierre Alliez, and Leif Kobbelt. Integer-grid maps for reliable quad meshing. *ACM Transactions on Graphics*, 32(4):1–12, 2013. doi: 10.1145/2461912.2462014. Proc. SIGGRAPH 2013.

[10] David Bommes, Bruno Lévy, Nico Pietroni, Enrico Puppo, Claudio Silva, Marco Tarini, and Denis Zorin. Quad-mesh generation and processing: A survey. *Computer Graphics Forum*, 32(6):51–76, 2013. doi: 10.1111/cgf.12014.

[11] Marcel Campen. Partitioning surfaces into quadrilateral patches: A survey. *Computer Graphics Forum*, 36(8):567–588, 2017. doi: 10.1111/cgf.13153. Abstract read.

[12] Marcel Campen, David Bommes, and Leif Kobbelt. Dual loops meshing: Quality quad layouts on manifolds. *ACM Transactions on Graphics*, 31(4):110:1–110:11, 2012. doi: 10.1145/2185520.2185606. Proc. SIGGRAPH 2012. Abstract read.

[13] Marcel Campen, David Bommes, and Leif Kobbelt. Quantized global parametrization. *ACM Transactions on Graphics*, 34(6):1–12, 2015. doi: 10.1145/2816795.2818140. Proc. SIGGRAPH Asia 2015. Abstract read.

[14] Beifang Chen and Jue Wang. The flow and tension spaces and lattices of signed graphs. *European Journal of Combinatorics*, 30(1):263–279, 2009. Abstract read.

[15] Wei Chen, Xiaopeng Zheng, Jingyao Ke, Na Lei, Zhongxuan Luo, and Xianfeng Gu. Quadrilateral mesh generation I: Metric based method. *Computer Methods in Applied Mechanics and Engineering*, 356:652–668, 2019. doi: 10.1016/j.cma.2019.07.023. Theorems 3.7, 3.8 and Lemma 3.9 of the arXiv version read.

[16] John W. Chinneck and Erik W. Dravnieks. Locating minimal infeasible constraint sets in linear programs. *ORSA Journal on Computing*, 3(2):157–168, 1991. doi: 10.1287/ijoc.3.2.157. Abstract read.

[17] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021. doi: 10.1007/978-3-030-79876-5_37.

[18] Irit Dinur and Samuel Safra. On the hardness of approximating vertex cover. *Annals of Mathematics*, 162(1):439–485, 2005. doi: 10.4007/annals.2005.162.439. Abstract read.

[19] Irit Dinur, Subhash Khot, Guy Kindler, Dor Minzer, and Muli Safra. On non-optimally expanding sets in Grassmann graphs. In *Proceedings of the 50th Annual ACM Symposium on Theory of Computing (STOC 2018)*, pages 940–951, 2018. doi: 10.1145/3188745.3188806. Journal version: Israel Journal of Mathematics 243 (2021), 377–420. Not read.

[20] Irit Dinur, Subhash Khot, Guy Kindler, Dor Minzer, and Muli Safra. Towards a proof of the 2-to-1 games conjecture? In *Proceedings of the 50th Annual ACM Symposium on Theory of Computing (STOC 2018)*, pages 376–389, 2018. doi: 10.1145/3188745.3188804. Not read.

[21] Dust3D project. AutoRemesher: Automatic quad remeshing. `https://github.com/huxingyi/autoremesher`, 2026. Open-source software (MIT licence), the version installed on the bench machine in October 2026. Used as a baseline in the bench.

[22] Jack Edmonds and Ellis L. Johnson. Matching: A well-solved class of integer linear programs. In Richard Guy, Haim Hanani, Norbert Sauer, and Johanan Schönheim, editors, *Combinatorial Structures and Their Applications (Proc. Calgary International Conference, 1969)*, pages 89–92. Gordon and Breach, New York, 1970. Reprinted in Combinatorial Optimization — Eureka, You Shrink!, LNCS 2570, Springer, 2003, pp. 27–30, doi:10.1007/3-540-36478-1_3, whose first page was read.

[23] Ralph H. Fox. Free differential calculus. I: Derivation in the free group ring. *Annals of Mathematics*, 57(3):547–560, 1953. doi: 10.2307/1969736. Cited for the definition of the derivatives and their fundamental formula, which are standard.

[24] Michael R. Garey, David S. Johnson, and Larry Stockmeyer. Some simplified NP-complete graph problems. *Theoretical Computer Science*, 1(3):237–267, 1976. doi: 10.1016/0304-3975(76)90059-1. Abstract read.

[25] Johan Håstad. Clique is hard to approximate within $n^{1-\varepsilon}$. *Acta Mathematica*, 182(1): 105–142, 1999. doi: 10.1007/BF02392825. Abstract read.

[26] Martin Heistermann, Jethro Warnett, and David Bommes. Min-deviation-flow in bi-directed graphs for T-mesh quantization. *ACM Transactions on Graphics*, 42(4), 2023. doi: 10.1145/3592437. Proc. SIGGRAPH 2023. Objective (sum of absolute deviations, Eq. 2) and solvers read; our quantizer follows its formulation.

[27] Jingwei Huang, Yichao Zhou, Matthias Nießner, Jonathan Richard Shewchuk, and Leonidas J. Guibas. QuadriFlow: A scalable and robust method for quadrangulation. *Computer Graphics Forum*, 37(5):147–160, 2018. doi: 10.1111/cgf.13498. Proc. SGP 2018. Used as a baseline.

[28] Qi Huangfu and J. A. Julian Hall. Parallelizing the dual revised simplex method. *Mathematical Programming Computation*, 10(1):119–142, 2018. doi: 10.1007/s12532-017-0130-5. The usual citation for the HiGHS solver.

[29] Ivan Izmestiev, Robert B. Kusner, Günter Rote, Boris Springborn, and John M. Sullivan. There is no triangulation of the torus with vertex degrees 5, 6, ..., 6, 7 and related results: geometric proofs for combinatorial theorems. *Geometriae Dedicata*, 166(1):15–29, 2013. doi: 10.1007/s10711-012-9782-5. Read in full (arXiv v2): holonomy of the developing map, Burgers vector, no 3,5-quadrangulation of the torus.

[30] Imke Joormann and Marc E. Pfetsch. Complexity of minimum irreducible infeasible subsystem covers for flow networks. Optimization Online preprint 4809, `https://optimization-online.org/2015/03/4809/`, 2015. Abstract and introduction read; a journal version was not found.

[31] Tomáš Kaiser and Matěj Stehlík. Colouring quadrangulations of projective spaces. *Journal of Combinatorial Theory, Series B*, 113:1–17, 2015. doi: 10.1016/j.jctb.2014.12.007. Lemma 3.1 read (arXiv:1310.5875v2).

[32] Felix Kälberer, Matthias Nieser, and Konrad Polthier. QuadCover – surface parameterization using branched coverings. *Computer Graphics Forum*, 26(3):375–384, 2007. doi: 10.1111/j.1467-8659.2007.01060.x. Proc. Eurographics 2007. Abstract read.

[33] Felix Kälberer, Matthias Nieser, and Konrad Polthier. Stripe parameterization of tubular surfaces. In *Topological Methods in Data Analysis and Visualization: Theory, Algorithms, and Applications*, Mathematics and Visualization, pages 13–26. Springer, 2011. doi: 10.1007/978-3-642-15014-2_2. Preprint read in full (Sections 3–4: integer gaps along cut paths, rounded).

[34] Subhash Khot and Oded Regev. Vertex cover might be hard to approximate to within $2-\varepsilon$. *Journal of Computer and System Sciences*, 74(3):335–349, 2008. doi: 10.1016/j.jcss.2007.06.019. Abstract read.

[35] Subhash Khot, Dor Minzer, and Muli Safra. On independent sets, 2-to-2 games, and Grassmann graphs. In *Proceedings of the 49th Annual ACM Symposium on Theory of Computing (STOC 2017)*, pages 576–589, 2017. doi: 10.1145/3055399.3055432. Abstract read.

[36] Subhash Khot, Dor Minzer, and Muli Safra. Pseudorandom sets in Grassmann graph have near-perfect expansion. In *59th IEEE Annual Symposium on Foundations of Computer Science (FOCS 2018)*, pages 592–601, 2018. doi: 10.1109/FOCS.2018.00062. Journal version: Annals of Mathematics 198(1) (2023), 1–92, doi:10.4007/annals.2023.198.1.1. Abstract read.

[37] Maurice Kléman and Jacques Friedel. Disclinations, dislocations, and continuous defects: A reappraisal. *Reviews of Modern Physics*, 80(1):61–115, 2008. Section II.5.1, Eq. (5) read (arXiv:0704.3055).

[38] Youkang Kong, Yang Liu, Yue Dong, Xin Tong, and Heung-Yeung Shum. SQuadGen: Generating simple quad layouts via chart distance fields. arXiv:2604.27329, 2026. Section 4.2 read: loop simplicity metrics.

[39] Na Lei, Xiaopeng Zheng, Jian Jiang, Yu-Yao Lin, and David Xianfeng Gu. Quadrilateral and hexahedral mesh generation based on surface foliation theory. *Computer Methods in Applied Mechanics and Engineering*, 316:758–781, 2017. doi: 10.1016/j.cma.2016.09.044. Abstract read.

[40] Na Lei, Xiaopeng Zheng, Zhongxuan Luo, Feng Luo, and Xianfeng Gu. Quadrilateral mesh generation II: Meromorphic quartic differentials and Abel–Jacobi condition. *Computer Methods in Applied Mechanics and Engineering*, 366:112980, 2020. doi: 10.1016/j.cma.2020.112980. Main theorems read (Thm. 4.7–4.13 of the arXiv version).

[41] Na Lei, Yiming Zhu, Xiaopeng Zheng, Hang Si, Zhongxuan Luo, and Xianfeng Gu. Why cross fields are not equivalent to quadrilateral meshes. *Computer Methods in Applied Mechanics and Engineering*, 417:116442, 2023. doi: 10.1016/j.cma.2023.116442. Abstract read.

[42] Arjen K. Lenstra, Hendrik W. Lenstra, Jr., and László Lovász. Factoring polynomials with rational coefficients. *Mathematische Annalen*, 261(4):515–534, 1982. doi: 10.1007/BF01457454. Bibliographic record checked.

[43] Jian Liu, Chunshi Wang, Song Guo, Haohan Weng, Zhen Zhou, Zhiqi Li, Jiaao Yu, Yiling Zhu, Jing Xu, Biwen Lei, Zhuo Chen, and Chunchao Guo. QuadGPT: Native quadrilateral mesh generation with autoregressive models. arXiv:2509.21420, 2025. Version 2 (February 2026). Appendix B.2 read: quad rings and quad lines as reward signals.

[44] George S. Lueker. Two NP-complete problems in nonnegative integer programming. Technical Report 178, Computer Science Laboratory, Department of Electrical Engineering, Princeton University, March 1975. As cited; not read.

[45] Max Lyon, Marcel Campen, and Leif Kobbelt. Quad layouts via constrained T-mesh quantization. *Computer Graphics Forum*, 40(2):305–314, 2021. doi: 10.1111/cgf.142634. Proc. Eurographics 2021. Read.

[46] Erick Moreno-Centeno and Richard M. Karp. The implicit hitting set approach to solve combinatorial optimization problems with an application to multigenome alignment. *Operations Research*, 61(2):453–468, 2013. doi: 10.1287/opre.1120.1139. Abstract read.

[47] Ashish Myles, Nico Pietroni, Denis Kovacs, and Denis Zorin. Feature-aligned T-meshes. *ACM Transactions on Graphics*, 29(4):1–11, 2010. doi: 10.1145/1778765.1778854. Proc. SIGGRAPH 2010. Cited for the remark of Bommes et al. (2011) about it; not read.

[48] Ashish Myles, Nico Pietroni, and Denis Zorin. Robust field-aligned global parametrization. *ACM Transactions on Graphics*, 33(4):135:1–135:14, 2014. doi: 10.1145/2601097.2601154. Proc. SIGGRAPH 2014. Abstract read.

[49] Mark Parker and Jennifer Ryan. Finding the minimum weight IIS cover of an infeasible system of linear inequalities. *Annals of Mathematics and Artificial Intelligence*, 17(1):107–126, 1996. As described by citing works; not read.

[50] Nico Pietroni, Enrico Puppo, Giorgio Marcias, Roberto Scopigno, and Paolo Cignoni. Tracing field-coherent quad layouts. *Computer Graphics Forum*, 35(7):485–496, 2016. doi: 10.1111/cgf.13045. Proc. Pacific Graphics 2016. Abstract read.

[51] Faniry H. Razafindrazaka and Konrad Polthier. Optimal base complexes for quadrilateral meshes. *Computer Aided Geometric Design*, 52–53:63–74, 2017. doi: 10.1016/j.cagd.2017.02.012. Proc. GMP 2017. Abstract and introduction read.

[52] Faniry H. Razafindrazaka, Ulrich Reitebuch, and Konrad Polthier. Perfect matching quad layouts for manifold meshes. *Computer Graphics Forum*, 34(5):219–228, 2015. doi: 10.1111/cgf.12710. Proc. SGP 2015. Abstract read.

[53] Kurt Strebel. *Quadratic Differentials*, volume 5 of *Ergebnisse der Mathematik und ihrer Grenzgebiete (3)*. Springer, Berlin, 1984. doi: 10.1007/978-3-662-02414-0. Not read; cited for the definition of Jenkins–Strebel differentials and the existence theorem with prescribed heights.

[54] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381, 2020. doi: 10.1145/3372885.3373824.

[55] William P. Thurston. Shapes of polyhedra and triangulations of the sphere. In *The Epstein Birthday Schrift*, volume 1 of *Geometry & Topology Monographs*, pages 511–549. Geometry & Topology Publications, Coventry, 1998. Abstract read (arXiv:math/9801088).

[56] Francesco Usai, Marco Livesu, Enrico Puppo, Marco Tarini, and Riccardo Scateni. Extraction of the quad layout of a triangle mesh guided by its curve skeleton. *ACM Transactions on Graphics*, 35(1):6:1–6:13, 2015. doi: 10.1145/2809785. Abstract read.

[57] William A. Veech. Flat surfaces. *American Journal of Mathematics*, 115(3):589–689, 1993. Bibliographic data checked; not read.

[58] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. doi: 10.1038/s41592-019-0686-2.

[59] Oleg V. Yazyev and Steven G. Louie. Topological defects in graphene: Dislocations and grain boundaries. *Physical Review B*, 81(19):195420, 2010. doi: 10.1103/PhysRevB.81.195420. Abstract read.

[60] Thomas Zaslavsky. Signed graphs. *Discrete Applied Mathematics*, 4(1):47–74, 1982. Bibliographic data checked; not read.

[61] Xiaopeng Zheng, Yiming Zhu, Wei Chen, Na Lei, Zhongxuan Luo, and Xianfeng Gu. Quadrilateral mesh generation III: Optimizing singularity configuration based on Abel–Jacobi theory. *Computer Methods in Applied Mechanics and Engineering*, 387:114146, 2021. doi: 10.1016/j.cma.2021.114146. Abstract read.

[62] David Zuckerman. Linear degree extractors and the inapproximability of max clique and chromatic number. *Theory of Computing*, 3(6):103–128, 2007. doi: 10.4086/toc.2007.v003a006. Abstract read.

A Provenance of the figures

The engine changed while the experiments were made. This appendix says, for each group of figures, which version of the engine produced it and where its raw output is. "Commit message" means that the figures are stated in the message of that commit of the repository; "working logs" means the output files and scripts of the experiment, kept by the author outside the repository and not yet published (Section 11, paragraph "Reproducibility").

figures	engine	source
twelve models, three budgets, 4.65% (Section 6)	the cited commit	working logs of the protocol below; a later run of the same engine gave the same 36 meshes' figures
Table 1	commits `7dc1727`, `c86740f`, `cf4a82c`	commit messages (also `ab8d89c`)
against other remeshers	commits `83f0967`, `4e9620c`	`bench/results/results.json` at those commits
hand-made male body	—	`bench/results/RESULTS.md`
male body, openclare	`ab8d89c`; `b0d37fa` to `8fd1932`	commit messages; closed loops, distance and corners: working logs
quad sphere	`18bc18c`	commit message and working logs
Sections 7, 8.2 to 8.4, 9; Appendices B, C, F	branch `rd-exact` (export hook), engine before `3b3122d`	working logs, scripts and the exported systems; some summary figures of Section 7 and Appendix F also in commit messages (`6374097`, `38b9fb3`)
Appendix G	branch `maxclose-rd`, engine of branch `complete-helices`	working logs; some figures only in the experiment's summary
Appendix D	a patch of the engine of commit `83f0967`	working logs; the ratios over twelve moves, the 36 runs and the 3,600 of 3,604 rings only in the experiment's summary
Section 8.5	not recorded (October 2026, before the cited commit)	working logs; the counts of sides (1,076, 314, 586) only in the experiment's summary

The protocol of the twelve models. For each model m (creature, blob, suzanne, spot, pig, ogre, bob, jerry, blub, rock with the preset `organic`; cage and part with `hard-surface`) and

each budget $Q \in \{2{,}970, 3{,}000, 3{,}030\}$: `openclare remesh bench/out/`$m$`/source.obj` *out.obj* `-preset` P `-quads` Q, then `openclare measure` on the source and the output with `-target` Q `-json`; the figures are the means of `loop_edges_spiral`, `loop_edges_closed`, `loops_spiral`, `mean_distance` and `angle_deviation_mean` over the 36 runs.

B Measured on the male body

The statements of Section 8 concern the T-mesh. We exported it with every turn, for the eight ways of the male body at 10,590 quads: its 1,295 nodes (1,034 T-junctions and 261 singular vertices, 133 of valence three, 127 of valence five and one of valence two, of total index 8) with their corner angles, its 2,069 arcs with their traces, its 776 patches, and for each turn the arcs it crosses. A turn is drawn as a curve by straight chords across the patches it crosses, and the regions its curve cuts the surface into are found from the chords. Identities between quantities of this appendix are identities between linear forms in the arc lengths modulo the balances, checked on a basis of the balanced lengths in floating point; parities are evaluated at the whole lengths the engine found. Counts are summed over the eight ways, a turn found in several ways counting once in each (measured throughout).

What holds. For each of the 1,299 turns, the crossing maps composed along the walk give a translation whose component across is the exported shift (Proposition 1). Of these turns, 718 have a simple curve; each splits the 261 singular vertices into two groups of total index 4 (Lemma 1). On each side of each of them, with a tree of cuts found by breadth-first search, the holonomy of the curve equals the composition of Proposition 3(a), in the order of first visit, in both components: 1,436 discs holding 4 to 257 singular vertices, no exception. The order matters: taking the vertices in the order of their last visit fails on 150 of the 158 discs of way 2. The checkerboard rule (Proposition 5) holds on the 1,436 discs, in the form given after Corollary 1, and on the 30,148 annuli between two disjoint simple turns, whose singular vertices have total index 0 in every case, as Lemma 1 says. The 294 pairs of disjoint turns with no singular vertex between them have equal shifts (Proposition 3(c)). Ten annuli hold exactly one vertex of valence three and one of valence five; on all ten the identity of Proposition 4(i) holds, and in none of them do the two vertices lie on one trace, so part (ii) never applies at the real lengths. The parity of Proposition 6 holds for the 1,299 turns.

Most held turns are not simple. Of the 1,299 turns, 101 cross themselves once, as in Lemma 2: 97 split the singular vertices as $(3, 3, 2)$, and 4 as $(2, 2, 4)$, all four crossing their closing segment. The other 480 cross themselves at least twice, up to 107 times. Of the 765 turns held for closing, only 356 are simple: long turns cross themselves many times, and the statements about simple curves and annuli say nothing about them. For such curves we use winding numbers: a combination $\sum_\gamma u_\gamma \bar{\gamma}$ of curves winds round each region of their union a whole number of times. This is its class in the homology of Σ°; it tells round which singular vertices the combination goes, but not its holonomy, which is not commutative.

C More on the conflicts of the male body

The larger sets. Section 8.2 describes the smaller sets. At the left hand (66, 71, 75, 76 of way 1, the same curves as 73, 78, 82, 83 of way 3), 76 is a figure-eight round three vertices of valence three, 29.9 long, and the others are 96 to 140 long, crossing themselves 13 to 28 times and each other 10 to 46 times; relative to the rest of the body, the weighted cycle winds round 18 of the 22 regions, with windings from $-\frac{1}{2}$ to $\frac{4}{3}$ in the normalization of Table 4. The set of seven in way 5 contains the two turns of configuration B and winds ± 4 round the two vertices they exchange; its other turns, up to 398 long, go round parts of both hands. The set of seven

in way 7, at the left foot, contains a turn of length 4.92 round a cap of four vertices of valence three, whose multiplier is 16 against ±1 or 2 for the six others, once scaled to whole numbers.

The conjecture in these terms. Since $\sum_i (I - R_i)^\top W_i = 0$, the vertices with non-zero twisted winding can be grouped so that $\sum_{i \in G} (I - R_i)^\top W_i = 0$ in each group G; each group then contributes a combination of separations of its own vertices. A group of a vertex of valence three and one of valence five contributes $\langle V, p_c - p_{c'} \rangle$ with $V = (I - R_c)^\top W_c$, a vector whose coordinates have equal parity, that is $a(d_1 + d_2) + b(d_1 - d_2)$ with a, b whole and $d = p_c - p_{c'}$: the diagonal forms of Proposition 4, which a staircase makes sign-definite (Proposition 7). An exchange is a group of two vertices of equal valence, a cap a group of total index four. So Question 2 says that the twisted windings of an irreducible certificate, in a suitable development, split into such groups, each with a form of the sign of the certificate. At the left hand, developed from the patch of turn 76, twelve vertices have a non-zero twisted winding. The vertices 231 (valence three) and 228 (valence five) have $W = (-5, -5)$ and $(-5, 5)$; their group contributes, modulo the balances, exactly $10\,\ell_{1805}$ to $-(7s_{66} + 5s_{71} + 2s_{75} + 3s_{76}) = 10\,\ell_{1805} + 2\ell_{1817} + 4\ell_{1853} + 2(\ell_{1870} + \ell_{1879} + \ell_{1969} + \ell_{1998} + \ell_{2029})$. A separatrix of 231 (trace 915, 7.76 edges) stops on a separatrix of 228 at the T-junction 1142, and arc 1805 joins that junction to 228, 0.063 edges long: ℓ_{1805} is the offset of the pair across the line of the field that nearly joins them, the case of Proposition 4(ii) with $\delta = \ell_{1805}$. The coefficient 5 of F is this dislocation, round which the weighted cycle goes five times in the twisted sense. In this chart the ten other vertices, six of valence three and four of valence five, form a group of total index 2 that no search could split. That group is an artefact of the development: the twisted windings are defined only up to the relation, and their support depends on the chart. Developed from another patch, eleven vertices have a non-zero twisted winding, among them 225, 234 and 236, and 229 is missing. The intrinsic reading is by combs (Section 8.4): the rest of the certificate is $2\ell_{1817} + 4\ell_{1853} + 2(\ell_{1870} + \ell_{1879} + \ell_{1969} + \ell_{1998} + \ell_{2029})$, eight comb readings on seven arcs forming a tree on the vertices 229, 235, 238, 239 (valence three) and 250 to 253 (valence five), of total index 0, the groups sharing vertices (measured). The sets of seven behave alike. In way 5, 24 vertices have a non-zero twisted winding in our chart; the pair 102, 103 exchanged in configuration B has $W = \pm(4, -4)$ and a form of the sign of the certificate, the pair 109, 118 of configuration C has $W = \pm(4, 0)$ and a form of the opposite sign; read by combs, the certificate is 13 comb readings on 11 vertices, whose odd part is a closed chain of three combs round 111, 112 and 128. In way 7, once the part of the words that is a multiple of the relation is removed, 16 vertices remain (141 to 151, 154 to 156, 165 and 169), and the certificate, with the primitive multipliers $(16, 1, 2, 1, 1, -1, 1)$, is $2(\ell_{1112} + \ell_{1165} + \ell_{1181} + \ell_{1201} + \ell_{1208} + \ell_{1232} + \ell_{1234})$: a tree of seven combs on the eight vertices 144 to 147, 150, 151, 155 and 156, of total index -2 (measured).

D Moving singularities in the parametrization

Proposition 3 suggests a lever before quantization: move the singular vertices so that the Burgers sum of each ring vanishes. We tested it on the seamless maps of the development models (measured throughout this appendix; the experiment is a patch of the engine, not part of it).

The Burgers sum on other models. For every ring of the seamless map that leans by at most 0.12, we computed the sum of Proposition 3(a) over the singular vertices of its smaller side, cut open along a tree of faces and ordered by first visit along its border. On seven closed surfaces of genus zero (blob, rock, creature, spot, pig, ogre, blub), 3,600 of the 3,604 rings that bound a disc are reproduced to $10^{-5}(1 + L)$, L the length of the ring; the reading fails only where a side is not a disc (borders on suzanne, handles on bob, part and cage). With the border and period terms of Proposition 15 it holds there too (Section 8.5).

Moving a singularity does not move the shift. The linearization $\delta s = \langle e_2, Q_i(I - R_i)\,\delta p_i\rangle$ assumes the other singular vertices keep their developed positions. They do not: on the blob at 3,000 quads, moving one pole of a cap one edge changes the shift of the cap ring (length 30.7, shift -1.065) by -0.13 with a local re-solve of the map, where the sum predicts -0.60; with a global re-solve it moved by $+0.14$, the other way, against a prediction of -0.58: the other poles of the cap move in the map and absorb the change. Over twelve moves per model, the measured change is 0.12, 0.03 and 0.01 of the predicted one on the blob, the creature and the pig. Rotating the field rigidly by $\alpha = |s|/L$ on the smaller side instead moves the shift by 26%, 36%, 46%, 64% and 87% of αL as the rotated region grows by 0, 5, 10, 20 and 40 rings of vertices, at almost no cost in distortion (0.3362 to 0.3371). By Proposition 3(c) the shift is a property of the whole cylinder: shearing a long cylinder is expensive in the least-squares map, displacing a cone point in it is not, so the map keeps the shift the field gives the cylinder and moves its cone points instead. The identity holds exactly; it is the developed positions of the cone points that the field sets, not their places on the surface.

A lever that fails. Cancelling the pairs of valences three and five of a twisted annulus would change the shift by its $\pm d_1 \pm d_2$ (Proposition 4); its effect on the spirals of the meshes was not recorded, and we do not report it. Rotating the field cylinder by cylinder to bring the shifts to zero reduced the shifts of the map but moved the twist onto rings of 90 to 200 edges, whose sensitivity grows with their length: over 36 runs (12 models, 3 sizes), 2 improved and 4 regressed, the mean gain coming from a single run, which we read as chaos rather than progress. Our reading, a conjecture: moving singularities would act through the Burgers sum only in a map whose scale is free to follow them, a cone metric; with a map of fixed scale the lever is the orientation of the field along whole cylinders, long rings must be handled jointly, and the conflicts are better attacked in quantization, where the staircases and dislocations of Section 8 live.

E Rotations of other orders

Rotations of order six. On triangle and hexagonal meshes the rotations are powers of $\zeta = e^{i\pi/3}$ and whole lengths develop nodes onto the Eisenstein integers; a vertex of valence k has index $6 - k$, in sixths of a turn, and the indices add up to 6χ. A turn walks along one of six directions u and comes back translated by $T = \alpha u + s\,\zeta u$: s counts the rows of the family of lines along u, α depends on the oblique basis chosen, and in the basis $(u, \zeta u)$ the strip has $2\alpha + s$ triangles per turn. The Burgers sum, the order of first visit and the Fox–Burgers formula hold unchanged. A pair of valences five and seven has $B = \pm\zeta^{\pm 1} d$, of length $|d|$, and the difference of the shifts is $\pm d_1$ or $\pm(d_1 + d_2)$ with $d = d_1 u + d_2 \zeta u$: the two turns close together only if the pair lies along one of the two other line directions, and a pair joined along a line parallel to the turns always forces a helix. Staircases have three directions, two that climb and one along which the form of the shift difference vanishes. There is no parity: $1 - \zeta$ is a unit. When every singular vertex has even valence the rotations have order three and the congruence modulo $(1-\omega)^2 = -3\omega$ (compare Lemma 11 of [29], on holonomy of order three) gives, on a closed sphere, $\alpha \equiv s \pmod 3$; every ring has a length divisible by three, as on subdivided octahedra (rings of $3n$ edges), while on subdivided icosahedra rings of $5n$ edges take every residue (measured on subdivided octahedra and icosahedra). Whether a rule modulo three holds with borders or handles is a conjecture we leave open.

Other rotation orders. Nothing in the Burgers sum, the order of first visit or the Fox–Burgers formula uses that R has order four: they hold for the rotations of order six of triangle and hexagonal meshes, for cone metrics of any angles, and for motions of space. What changes is parity. When every rotation is a power z^{j_i} of one rotation z, every Fox derivative is congruent

to its exponent sum modulo $1-z$, and $1-z^j$ to $j(1-z)$ modulo $(1-z)^2$, so

$$\tau(w) \equiv (1-z)\sum_i \epsilon\Big(\frac{\partial w}{\partial x_i}\Big) j_i\, p_i \pmod{(1-z)^2},$$

proved in any commutative ring. For quarter-turns, $(1-z)^2 = -2z$, and this is Corollary 2, with $j_i = 4 - k(c_i)$. For rotations of order three, $(1-z)^2 = -3z$: a rule modulo three, with three colours. For rotations of order six, $1-z$ is a unit: there is no colouring rule on triangle or hexagonal meshes. A dislocation of order n has $|B| = 2\sin(\pi/n)\,|d|$: $\sqrt{2}\,|d|$ for quads, $|d|$ for a pair of valences five and seven on a triangle mesh, $\sqrt{3}\,|d|$ for rotations of order three; the forms $\pm d_1 \pm d_2$ are particular to quads. Discrete Gauss–Bonnet reads $\sum(4-k) = 4\chi$ on quad meshes, $\sum(6-k) = 6\chi$ on triangle meshes and $\sum(3-k) = 3\chi$ on hexagonal ones. Theorem 2, Corollary 3 and Lemma 5 do not depend on the geometry at all.

F Whole lengths

This appendix returns to the male body of Section 7, now in whole numbers. The propositions are proved; the numbers are measured, with exact integer or rational arithmetic where we say so, and otherwise from HiGHS through SciPy, stopped at a time limit (best solution found and proven bound).

Exact whole lengths. Closing exactly, in whole numbers, the turns found closable together with the bounds kept is feasible: HiGHS solves it in about two minutes a way for three of the four ways tried (the fourth, with 87 turns, did not finish within ten minutes). The solutions leave 1 to 4 held turns shifted where the engine leaves 13 to 44, and fewer of the turns not held as well (5 against 9, 5 against 11, 0 against 27), at a deviation from the real lengths 4% to 62% higher. Yet the quad meshes they make spiral *more*: 12.2% of their edges against 8.8% for the first way, 3.7% against 1.1% for the second, and the third breaks the structure. Most of the loops that spiral are thus not turns our search finds: closing every turn it finds is not enough.

All the strips of a mesh. Walking every strip of the mesh the engine of the export makes for way 2 (Proposition 2) finds 230 ways round; each strip is found twice, once in each direction, with opposite shift expressions (the surface being oriented), and 168 of them come back shifted. A selection of rings by the field's shift alone (under half a row) is too narrow: the twisted thigh, shifted 1.65 rows a turn by the field, is a ring a modeller wants closed. Selecting rings by their lean, as the constraint rounds of the engine now do (Section 4), gives 222 of the 230 ways round; 218 of them (109 strips) can be closed together in real numbers with every bound of separation kept. Posed directly, HiGHS found no whole solution within ten minutes, nor in a later run of thirty. The rest of this appendix explains why, and what that set of rings really costs.

The balances form a bidirected graph. The matrix B has two rows per patch, one per direction, 1,552 rows for the 776 patches, and one column per arc. On a closed surface where no patch borders itself across an arc, an arc lies on a side of each of the two patches it borders, so its column has exactly two non-zero entries, each $+1$ or -1 according to the side of the patch it lies on (measured on all eight systems; in way 2, 958 arcs have signs $+/-$, 557 have $-/-$ and 554 have $+/+$). In the language of signed graphs, B is the incidence matrix of a *bidirected graph* G whose nodes are the balance equations and whose edges are the arcs; an arc on a border of the surface has a single entry, a half-edge of G. Bidirected graphs are the setting of [26]; the rank of their incidence matrices and the circuits of the associated matroid are due to Zaslavsky [60], and integer flows are generated by circuits [14]. Proposition 18 below makes a lattice basis explicit. Call a cycle of G *balanced* if values ± 1 on its edges have a zero balance at each of its nodes, and *unbalanced* otherwise.

Lemma 5 (Handshake)**.** *Let $x \in \mathbb{Z}^n$ and C be a connected component of G. Then $\sum_{r\in C}(Bx)_r \equiv \sum_a x_a \pmod 2$, the second sum running over the border arcs of C. On a closed surface, $\sum_r (Bx)_r$ is even.*

Proof. It is $\sum_a x_a \sum_{r\in C} B_{ra}$, and each column sums to 0 or ± 2, or to ± 1 for a border arc. □

So on a closed surface the rows of B are dependent modulo 2, although they are independent over the reals: a system $Bx = d$ with $\sum_r d_r$ odd has real solutions but no whole one; and, G being connected with an unbalanced cycle, one with an even sum always has whole solutions (on the edges T of Proposition 18, the whole image of B is exactly the lattice of vectors of even sum; Section 10). Consistently, integer elimination of B ends on the male body with one row whose entries have greatest common divisor 2, all other pivots being units (measured in all eight ways). In general the number of such rows is the number of components of G with an unbalanced cycle and no half-edge: none on Suzanne and Jerry, whose borders give half-edges, one on Bob, part, cage and a cylinder (measured). This does not affect the balanced lengths:

Proposition 18 (Free arcs)**.** *Let G be connected and contain an unbalanced cycle. Let T be the edges of a spanning tree of G and one more edge closing an unbalanced cycle, and F the other edges. Then B has full row rank, and every $x_F \in \mathbb{Z}^F$ extends to exactly one $x \in \mathbb{Z}^n$ with $Bx = 0$. The whole balanced lengths thus form a lattice isomorphic to $\mathbb{Z}^F$, of dimension $n - m$ for m balances, with a basis of vectors z_f, $f \in F$, each supported on $T \cup \{f\}$ with $z_f(f) = 1$ and entries in $\{0, \pm 1, \pm 2\}$.*

Proof. Such a T exists: the sign of a cycle is a homomorphism from the cycle space of G over $\mathbb{Z}/2$, of which the fundamental cycles of a spanning tree are a basis, so if all of them were balanced every cycle would be. T has m edges. If $B_T y = 0$, a node met by a single edge of T forces that edge to zero; removing such edges repeatedly leaves the unbalanced cycle, on which each value fixes the next up to sign, and going round, since the cycle is unbalanced, forces them all to zero. So the square matrix B_T is invertible, B has full row rank, and x_F has exactly one real extension; it remains to see that it is whole. For $f \in F$, the graph $T \cup \{f\}$ has a one-dimensional space of balanced values. Removing the edges met at a node of degree one leaves either three paths between two nodes, whose three cycles are the unbalanced cycle of T, one more unbalanced and one balanced (the number of unbalanced cycles of such a graph is even), or two cycles joined by a path, possibly of length zero. In the first case the balanced cycle contains f, since the cycle of T does not, and carries values ± 1. In the second, the cycle through f either is balanced and carries values ± 1, or is unbalanced, and then values ± 1 on both cycles and ± 2 on the joining path are balanced. Either way there is a whole balanced z_f supported on $T \cup \{f\}$ with $z_f(f) = \pm 1$; changing its sign if needed, $x = \sum_{f\in F} x_f z_f$ is the whole extension. □

The hypothesis can fail beyond the closed sphere: on a torus without singular vertex and on part, B does not have full row rank (4 of 6 and 145 of 148, measured). The proposition extends with a spanning forest of G and, in each component with an unbalanced cycle, one edge closing such a cycle or one half-edge; the whole balanced lengths then form a lattice of dimension $n - m$ plus the number of balanced components without half-edge.

On the male body G is connected with an unbalanced cycle, B has rank 1,552, and the lattice of balanced lengths has dimension $2{,}069 - 1{,}552 = 517$ (measured in all eight ways). Balances alone thus never obstruct whole lengths: 517 arcs can be given any whole lengths and the others follow. With the shifts of the 218 rings of way 2 the rank grows by 105 and the lattice of closed quantizations has dimension 412; integer elimination still uses unit pivots only, with one row of greatest common divisor 2 (the shift of one ring is even as soon as the balances and the rings before it hold). Adding the shifts of the held turns instead gives dimensions 469, 439, 465, 431, 435, 363 and 431 in ways 0 to 6; in ways 4, 5 and 6 elimination needs steps of Euclid's algorithm, so turn rows can destroy the free-arc structure.

Integer solutions exist. By Remark 5, whole lengths closing the 218 rings within the bounds exist. HiGHS finds some in one second once it is told which arcs are zero in *every* real solution: closing the 218 rings with $\ell \geq 0$ forces 171 arcs to zero (measured), the implicit equalities of the cone, each certified by a non-negative combination of shifts as in Remark 4. There are 23 distinct certificates: 13 single rings whose shift is sign-definite (round fingers and toes), 4 pairs with multipliers $(1, 1)$ (left foot, right foot, head, top of the head), and 6 combinations of four to six rings (torso, neck). Given as upper bounds, these zeros make the problem easy; left implicit, the solver works on a face of lower dimension than the problem without knowing it, which is what the ten- and thirty-minute failures were. They are not an integer obstruction.

The difficulty is their size. The 171 arcs forced to zero carry 350.8 quad edges of target length, the longest 10.1. The real lengths closing the 218 rings nearest the targets, in the weighted deviation, make 3,960 quads instead of the 10,496 of the targets. With the deviation the engine measures, the mean of $|\ell_a - \lambda r_a|$, a linear program shows that no real lengths closing the 218 rings have a mean deviation below 0.817, nor, after dropping 7 strips (14 ways round) so that every arc with a target of at least 1.5 stays open, below 0.758; balanced lengths without rings reach 0.085. A lower bound in real numbers holds for whole numbers too. The engine version that made the export refused quantizations whose deviation exceeded 1.5 times that of the plain search, $1.5 \times 0.316 = 0.474$; the engine at the cited commit allows $\max(1.5 \times 0.316,\ 0.316 + 0.2) = 0.516$. Either way the 218 rings cannot close at the size asked for within the engine's tolerance, whatever the rounding: the obstruction is in the reals.

The 190 rings that fit. A subset of 190 of the 218 ways round (chosen greedily, so 190 is only a lower bound on what fits) closes in real numbers at a mean deviation of 0.399, keeping the number of quads (10,502) at that of the targets to first order: the linear form $\nabla q(t) \cdot \ell \geq \nabla q(t) \cdot t$, where $q(\ell)$ is the number of quads, a quadratic function of the lengths, and t the targets. In whole numbers we followed the reformulation of Aardal, Hurkens and Lenstra [1]: a basis M of the lattice of closed quantizations, with the 74 implicit zeros of this set imposed first (dimension 407), reduced by LLL [42] for a weighted inner product, which leaves entries in $[-2, 2]$; then $\ell = My$ with y whole, so that no equation is left, and a mixed-integer program over y. Babai's nearest plane [5] from the real solution gives a mean deviation of 0.586 but 10 arcs below their bounds. HiGHS, stopped after 1,200 seconds, gives whole lengths closing all 190 rings within every bound, with a mean deviation of 0.619 and 10,568 quads, and proves that no whole lengths do better than 0.536 under the same constraints, the first-order proxy of the number of quads included. Against 0.399 in real numbers, this rounding gap is the one integer obstruction we found, and it is not a parity condition: every infeasible set of Table 4 is infeasible in real numbers already.

Which turns, not rounding. Given to the engine in place of the lengths of way 2, these whole lengths close all 190 rings, with 10,568 quads and 2.5% of spiral edges against 1.1% for the engine's own lengths (the share of that way alone: the deviation cap would refuse it). At the new lengths the strips are different walks: walking all of them finds 230 ways round, 156 of them shifted (168 at the engine's lengths), and of the 214 that lean less than 0.12, 142 are shifted (162 of 222 at the engine's lengths). Closing the rings of one set of lengths moves the strips, and other loops spiral. On this body, then, whole lengths are not what is missing: rounding costs between 0.137 (proven) and 0.220 (found) of mean deviation above the real optimum, and it is the choice of the turns to close, at the lengths where they are the strips of the mesh, that decides how much the mesh spirals.

G Closing the most rings, end to end

Most rings closable, end to end. Hardness is a statement about the worst case; we also measured what an exact selection buys on the bench, with the engine of branch `complete-helices` (the constraint rounds without the finer work mesh of the cited commit: 4.54% of spiral edges over the 36 runs, against 4.65%). We posed, for the lengths the engine keeps, the mixed-integer program that closes as many of its strips leaning less than 0.12 as it can (weighted by their length round), within the balances, the bounds of separation, the engine's cap on the mean deviation and its quad budget, and meshed the result with the engine itself; then walked the strips at the new lengths and solved again, three rounds. Posed with the shift expressions alone, the program closes 27.3 of the 38.2 such strips of a model (12 models, three budgets each), but at the new lengths only 22.1 are closed strips, against 21.8 for the engine: the strips have become other walks. Each walk stays the same where the place it crosses every arc stays inside that arc, half a quad edge from either end, two linear inequalities per crossing; required for the rings counted as closed, they make the count exact (24.6 closed by the program, 24.0 found closed by walking the mesh) and show that, within the engine's cap and budget, it closes only about three strips fewer than can be closed together without changing walk (measured).

What it buys. In place of the engine's lengths, these lengths do not spiral less: 4.77% and 5.08% of spiral edges in two runs against 4.54%, better on 13 and 10 runs of 36, worse on 10 and 13. Added to the lengths the engine chooses among, by its own count of spirals (which equals the measured share on every run checked), they lower the mean to 2.32% and 2.70%, never higher. Three other draws of the engine's per-arc preferences, added the same way, give 4.12%, and three lengths drawn at random within the same cap and budget 4.34%; without the conditions on the walks, the rounds give 3.15%, no better than the other draws (better on 7 runs, worse on 7). The other measures barely move (closed loops 38.0% against 36.3% of edges, mean distance 0.383 against 0.385, all 36 meshes valid). On the male body every round spiralled more than the engine (from 3.45% to 5.59% against 1.59%); there, within the time allowed, the program closed only 35 to 38 of 107 strips (the engine 33), and in 600 seconds at most 48 that keep their walk, 65 that do not, its bound remaining far above (measured). Closing the most rings is thus not, on its own, the way to fewer helices: the shifted strips the measure does not see as spirals are among those closed, and others take their place; it is a source of good candidates, valuable only where the strips closed stay the strips of the mesh.

Only when the solver finishes. The engine can afford a few seconds per solve. With a limit of 5 seconds the gain vanishes (an outcome that depends on the hardware and its load, Section 6; the distribution of solve times was not recorded): the engine with three rounds as candidates gives 4.10%, against 4.12% for three other draws (run by run, 2 better and 10 worse than those draws); 63 of the 126 pieces get no solution in 5 seconds, 21 more stop at the limit, and 68 of 108 rounds return the engine's own lengths (measured). A local search inside the engine, walks of its strip network biased by Lagrange multipliers with the conditions on the walks kept, costs 0.1 to 0.25 seconds per model but closes no more rings than the engine (6 of 30 on the blob, 12 of 26 on the creature, unchanged): a single walk almost always changes the shift of a ring already closed, and closing needs combinations of walks. The gain comes from a solver that finishes, and Theorem 4 says that no exact method finishes fast on every instance.

H What Lean checks, statement by statement

- For motions whose linear parts lie in any ring acting on any module: the translation of $\rho_m \circ \cdots \circ \rho_1$ is $\sum_i Q_i(I - R_i)p_i$ with $Q_i = R_m \cdots R_{i+1}$ (Proposition 3(a)); the recursion of the gluing maps of a tree of cuts unfolds into the order of first visit, and the order of

last visit gives another vector already for two vertices; the holonomy of the border of a cut annulus (Proposition 3(b)); exchanges (Proposition 8); the conjugation of Proposition 9; the additivity behind Proposition 1.

- Over the reals: the four orders and orientations of a dislocation, the forms $\pm d_1 \pm d_2$ and $|B_A|^2 = 2|d|^2$, straight chains (Proposition 4(i), (ii)); staircases (Proposition 7); the positivity of a non-negative form with a held arc (Remark 4).

- At whole lengths: the parities of $(I - R^{\pm 1})p$ and $(I - R^2)p$; the checkerboard rule (Proposition 5); the induction of the proof of Proposition 6, given a 2-colouring of the quad graph (the extension from positive to all whole lengths is on paper), and the second proof of Remark 2.

- The Fox–Burgers formula on words, which is well defined on the free group, the fundamental formula, the twisted windings and $\sum_i (I - R_i)^\top W_i = 0$, the insertion of the relation (Proposition 10); the commutator (Proposition 11); Corollary 2, with the two arguments of its proof on the colour of the origin and the constant of the winding numbers.

- Lemma 5 on a closed surface, Theorem 2 and Corollary 3.

- Proposition 18, with switchings and spanning trees grown by leaves, and stronger: on its edges T, the whole image of B is the lattice of vectors of even sum.

- Remarks 4 and 5: Farkas' lemma by Fourier–Motzkin elimination over any ordered field, the arcs forced to zero, whole multipliers and whole solutions.

- Proposition 2, in a model only: there the strips are defined as the cycles of a successor map built from a permutation of the half-integer stations, so the bijection holds by construction; separately, the real walk from $k + \frac{1}{2}$ follows the unit edges in a one-dimensional model of the gluing. That the station walk of the gluing data is a permutation and that the model is the quad mesh are assumptions.

- The Euler-characteristic count used in Lemmas 1 and 2, on the grid refined by two.

Not checked: the topology the discrete statements take as hypotheses (that a simple closed curve bounds a disc, the developing map and the presentation of $\pi_1(\Sigma^\circ)$), which corners of the regions of Lemma 2 are convex, strips on surfaces with borders, and every measured statement.

Invariants for the engine. The checked identities give tests on the engine's own quantities; since commit `3bcbd20` its debug builds check the first three below, and it uses the parity and the sign-definite turns to spare work (Section 4): on a closed surface of genus zero, the two components of the translation of every turn, its length and its shift, have the same parity at the whole lengths of the integer search, so every ring has an even number of quads round it; the coefficients of every shift expression are whole; every column of the balances has two entries ± 1, or one for an arc on a border, so on a closed surface a balance system with an odd total right-hand side has no whole solution, and in general the sum over a component has the parity of the sum of the values on its border arcs; where the developed cone points are known, the translation of a turn bounding a disc is the Burgers sum in the order of first visit, and its two components have the parity of the number of vertices of odd valence on points of odd colour; a turn whose shift is a non-negative form with a held arc, or both turns of a staircase with a held arc, are never closed; and the number of held turns left shifted is at least the number of disjoint infeasible sets found. The README of the folder lists them with the engine's names.